

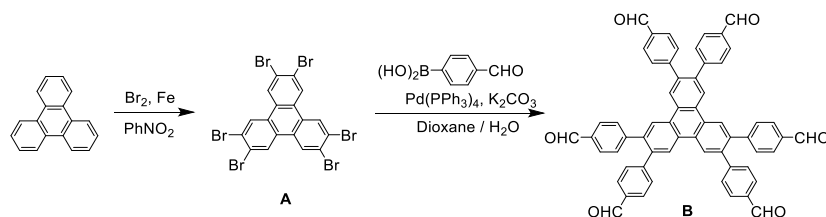
Linkage-engineered donor–acceptor covalent organic frameworks for optimal photosynthesis of hydrogen peroxide from water and air

In the format provided by the authors and unedited

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Supplementary Methods

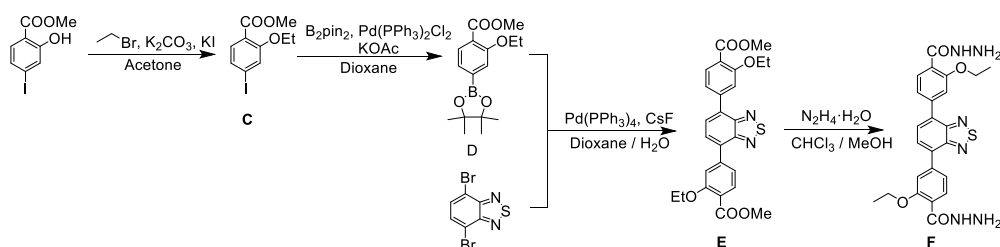
Synthesis of 2,3,6,7,10,11-hexakis(4-formylphenyl) triphenylene (B)



2,3,6,7,10,11-Hexabromotriphenylene (A): To a mixture of triphenylene (4.38 mmol, 1 g) and iron powder (1.79 mmol, 0.1 g) in nitrobenzene (40 mL), bromine (40 mmol, 2 mL) was added dropwise over 10 min. The solution was stirred at room temperature overnight and then heated in sand bath at 200 °C for 2 h. After cooling to room temperature, the mixture was poured into diethyl ether (150 mL) and filtered. The filter cake was washed with diethyl ether (50 mL $\times$ 3) and then recrystallised from 1,2-dichlorobenzene to afford A as a white solid in 90% yield. A was used directly in the next step.

2,3,6,7,10,11-Hexakis(4-formylphenyl) triphenylene (B): A two-neck round bottom flask (250 mL) with 2,3,6,7,10,11-hexabromotriphenylene (1 mmol, 0.702 g), 4-formylphenylboronic acid (9 mmol, 1.35 g), potassium carbonate (30 mmol, 4.15 g) and dioxane/H₂O (75 mL/15 mL) was purged with argon for 10 min. Then tetrakis(triphenylphosphine)palladium (0.18 mmol, 0.208 g) was added under argon protection. The mixture was degassed via three freeze-pump-thaw cycles and heated under reflux at 100 °C for 3 days. After cooling to room temperature, solvents were removed by evaporation and the residual solid was extracted with chloroform (50 mL $\times$ 3) and washed with brine (50 mL). The combined organic layer was dried over anhydrous sodium sulphate and then subjected to column chromatography (DCM/EA = 20/1 v/v) to obtain B as a white solid in 92% yield. ¹H NMR (500 MHz, CDCl₃): δ (ppm) 10.03 (s, 6H), 8.76 (s, 6H), 7.84 (d, 12H, J = 6.8 Hz), 7.49 (d, 12H, J = 6.8 Hz). ¹³C NMR (126 MHz, CDCl₃, ppm): 191.65, 146.77, 139.34, 135.18, 130.66, 129.73, 129.44, 125.84 (Supplementary Fig. 43 and 50).

Synthesis of 4,4'-(benzo[*c*][1,2,5]thiadiazole-4,7-diyl)bis(2-ethoxybenzohydrazide) (F)



Methyl 2-ethoxy-4-iodobenzoate (C): To a mixture of methyl 2-hydroxy-4-iodobenzoate (20 mmol, 5.56 g), potassium carbonate (80 mmol, 11.2 g) and potassium iodide (1.2 mmol, 200 mg) in acetone (300 mL), bromoethane (40 mmol, 3 mL) was added dropwise. The mixture was heated under reflux at 65 °C overnight. After cooling to room temperature, the mixture was filtered through a Celite bed. The filtrate was evaporated to give a colourless oil product C in 98% yield. ¹H NMR (500 MHz, CDCl₃): δ (ppm) 7.49 (dd, J = 8.4, 1.4 Hz, 1H), 7.33 (dd, J = 8.2, 1.6 Hz, 1H), 7.31 (d, J = 1.5 Hz, 1H), 4.10 (q, J = 7.0 Hz, 2H), 3.88 (s, 3H), 1.46 (t, J

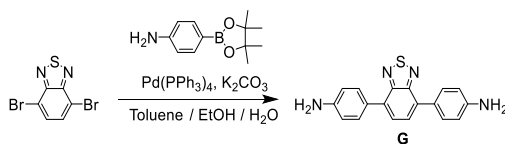
= 7.0 Hz, 3H). ¹³C NMR (126 MHz, CDCl₃, ppm): 166.22, 158.68, 132.69, 129.41, 122.74, 119.91, 99.84, 65.01, 52.04, 14.61 (Supplementary Fig. 44).

Methyl 2-ethoxy-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)benzoate (D): The above oil product C was added to a mixture of bis(pinacolato)diboron (21 mmol, 5.34 g), potassium acetate (60 mmol, 5.88 g) and Pd(PPh₃)₂Cl₂ (0.3 mmol; 0.21g) in 1,4-dioxane (100 mL) under argon. The mixture was heated at 100 °C for 24 h. After cooling to room temperature, the resulting mixture was filtered through a Celite bed to remove the inorganic salt. The filtrate was evaporated and then subjected to column chromatography (hexane/EA = 50/1 v/v) to obtain a colourless oil product D in 85% yield. ¹H NMR (500 MHz, CDCl₃): δ (ppm) 7.75 (d, *J* = 7.6 Hz, 1H), 7.41 (dd, *J* = 7.6, 1.0 Hz, 1H), 7.38 (s, 1H), 4.19 (q, *J* = 7.0 Hz, 2H), 3.90 (s, 3H), 1.47 (t, *J* = 7.0 Hz, 3H), 1.36 (s, 12H). ¹³C NMR (126 MHz, CDCl₃, ppm): 166.95, 157.70, 130.58, 126.39, 122.91, 119.09, 118.42, 84.16, 64.69, 51.95, 24.87, 14.78 (Supplementary Fig. 45).

Dimethyl 4,4'-(benzo[c][1,2,5]thiadiazole-4,7-diyl)bis(2-ethoxybenzoate) (E): A two-neck round bottom flask (100 mL) charged with 4,7-dibromobenzo[c]-1,2,5-thiadiazole (3.4 mmol, 1.0 g), oil product C (10.1 mmol, 3.1 g), cesium fluoride (20 mmol, 3.02 g) and dioxane/H₂O (30 mL/15 mL) was purged with argon for 10 min. Then tetrakis(triphenylphosphine)palladium Pd(PPh₃)₄ (0.3 mmol, 0.347 g) was added under argon protection. The mixture was degassed via three freeze-pump-thaw cycles and heated under reflux at 100 °C for 3 days. After cooling to room temperature, solvents were removed by evaporation and the residual solid was extracted with chloroform (50 mL × 3) and washed with brine (50 mL). The combined organic layer was dried over anhydrous sodium sulphate and then subjected to column chromatography (hexane/EA = 4/1 v/v) to obtain a bright yellow product E in 70% yield. ¹H NMR (500 MHz, CDCl₃): δ (ppm) 7.99 (d, *J* = 8.0 Hz, 2H), 7.87 (s, 2H), 7.71 (d, *J* = 1.5 Hz, 2H), 7.55 (dd, *J* = 8.0, 1.6 Hz, 2H), 4.28 (q, *J* = 7.0 Hz, 4H), 3.96 (s, 6H), 1.55 (t, *J* = 7.0 Hz, 6H). ¹³C NMR (126 MHz, CDCl₃, ppm): 166.57, 158.68, 153.84, 142.11, 132.99, 131.90, 128.33, 120.78, 120.17, 114.71, 77.28, 77.03, 76.78, 64.90, 52.04, 14.77 (Supplementary Fig. 46).

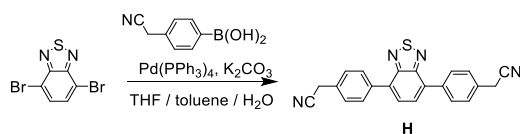
4,4'-(Benzo[c][1,2,5]thiadiazole-4,7-diyl)bis(2-ethoxybenzohydrazide) (F): To a mixture of bright yellow product E (0.61 mmol, 0.300 g) in methanol/chloroform (50 mL, 4/1 v/v), hydrazine monohydrate (5 mL) was added dropwise. The mixture was heated under reflux at 70 °C for 24 h. After cooling to room temperature, the precipitate was filtered and washed by methanol (20 mL × 3) to give a bright yellow product F in 95% yield. ¹H NMR (500 MHz, *d*₆-DMSO): δ (ppm) 9.24 (s, 2H), 8.08 (s, 2H), 7.87 (d, 2H, *J* = 8.0 Hz), 7.76 (d, 2H, *J* = 1.6 Hz), 7.68 (dd, 2H, *J* = 8.0, 1.6 Hz), 4.62 (s, 4H), 4.30 (q, 4H, *J* = 6.9 Hz), 1.44 (t, 6H, *J* = 7.0 Hz). ¹³C NMR (126 MHz, CDCl₃, ppm): 164.50, 155.95, 153.19, 140.16, 131.92, 130.35, 128.69, 122.15, 121.25, 113.70, 64.35, 14.47 (Supplementary Fig. 47 and 51).

Synthesis of 4,4'-(benzo[c][1,2,5]thiadiazole-4,7-diyl)dianiline (G)



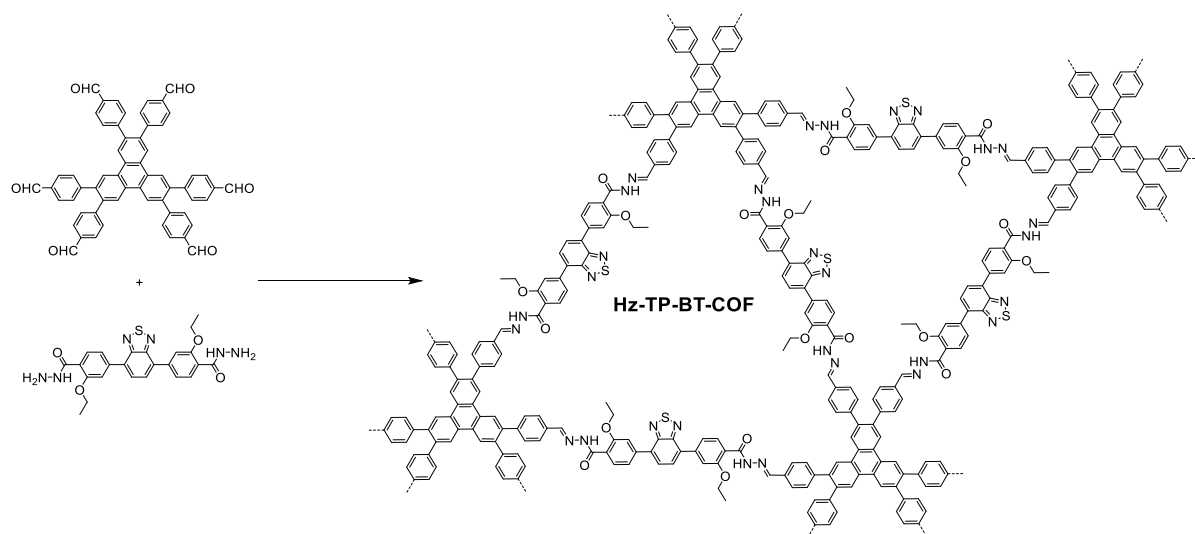
4,4'-(Benzo[c][1,2,5]thiadiazole-4,7-diyl)dianiline (G): A two-neck round bottom flask (100 mL) charged with 4,7-dibromobenzo[c]-1,2,5-thiadiazole (3.4 mmol, 1.0 g), 4-(4,4,5,5-tetramethyl-1,3,2)dioxaborolan-2-yl)aniline (10.1 mmol, 2.2 g), potassium carbonate (20 mmol, 2.76 g) and toluene/ethanol/H₂O (40 mL/20 mL/10 mL) was purged with argon for 10 min. Then tetrakis(triphenylphosphine)palladium (0.3 mmol, 0.347 g) was added under argon protection. The mixture was degassed via three freeze-pump-thaw cycles and heated under reflux at 100 °C for 3 days. After cooling to room temperature, solvents were removed by evaporation and the residual solid was extracted with chloroform (50 mL × 3) and washed with brine (50 mL). The combined organic layer was dried over anhydrous sodium sulphate and then subjected to column chromatography (DCM) to obtain a red product G in 70% yield. ¹H NMR (500 MHz, *d*₆-DMSO): δ (ppm) 7.76 (d, 4H, *J* = 8.5 Hz), 7.71 (s, 2H), 6.71 (d, 4H, *J* = 8.6 Hz), 5.40 (s, 4H). ¹³C NMR (126 MHz, *d*₆-DMSO, ppm): 153.65, 148.95, 130.97, 129.73, 126.12, 124.32, 113.69 (Supplementary Fig. 48 and 52).

Synthesis of 2,2'-(benzo[c][1,2,5]thiadiazole-4,7-diylbis(4,1-phenylene))diacetonitrile (H)



2,2'-(Benzo[c][1,2,5]thiadiazole-4,7-diylbis(4,1-phenylene))diacetonitrile (H): A two-neck round bottom flask (100 mL) charged with 4,7-dibromobenzo[c]-1,2,5-thiadiazole (1 mmol, 0.294 g), 4-(cyanomethyl)benzeneboronic acid (3 mmol, 0.747 g), potassium carbonate (10 mmol, 1.38 g) and THF/toluene/H₂O (40 mL/5 mL/5 mL) was purged with argon for 10 min. Then tetrakis(triphenylphosphine)palladium Pd(PPh₃)₄ (0.08 mmol, 0.092 g) was added under argon. The mixture was degassed via three freeze-pump-thaw cycles and heated under reflux at 85 °C for 3 days. After cooling to room temperature, solvents were removed by evaporation and the residual solid was extracted with chloroform (50 mL × 3) and washed with brine (50 mL). The combined organic layer was dried over anhydrous sodium sulphate and subjected to column chromatography (hexane/DCM = 2/1 v/v) to obtain a pale-yellow solid H in 82% yield. ¹H NMR (500 MHz, CDCl₃): δ (ppm) 8.00 (d, 4H, *J* = 8.3 Hz), 7.81 (s, 2H), 7.53 (d, 4H, *J* = 8.4 Hz), 3.87 (s, 4H). ¹³C NMR (126 MHz, CDCl₃, ppm): 153.94, 137.19, 132.74, 130.07, 129.98, 128.26, 128.11, 117.63, 23.50 (Supplementary Fig. 49 and 53).

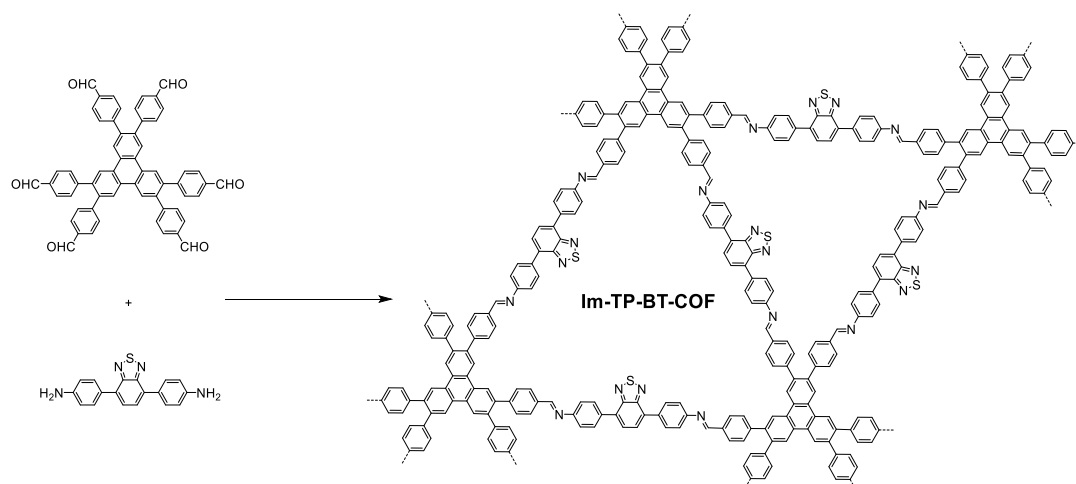
Synthesis of Hz-TP-BT-COF



Hz-TP-BT-COF: A 1,2-dichlorobenzene/acetonitrile (*o*-DCB/ACN = 0.7 mL/0.3 mL) mixture of 2,3,6,7,10,11-hexakis(4-formylphenyl) triphenylene (0.01 mmol, 8.5 mg) and 4,4'-(benzo[*c*][1,2,5]thiadiazole-4,7-diyl)bis(2-ethoxybenzohydrazide) (0.03 mmol, 14.8 mg) in a Pyrex tube (10 mL) was sonicated for 3 minutes and followed by addition of acetic acid (6 M, 0.1 mL). The mixture was then degassed via three freeze-pump-thaw cycles. The tube was sealed and heated at 150 °C for 3 days. The precipitate was collected by suction filtration, washed with THF three times, and then Soxhlet extracted by THF overnight. The powder was dried under vacuum at 120 °C to afford Hz-TP-BT-COF with yellow colour in an isolated yield of 86%. Elemental analysis for Hz-TP-BT-COF: Calculated: C, 78.64; H, 4.99; N, 7.64; S, 2.92%. Found C, 64.25; H, 4.65; N, 7.74; S, 3.44%.

Amorphous Hz-TP-BT polymer: A 1,2-dichlorobenzene/tetrahydrofuran (*o*-DCB/THF = 0.5 mL/0.5 mL) mixture of 2,3,6,7,10,11-hexakis(4-formylphenyl) triphenylene (0.01 mmol, 8.5 mg) and 4,4'-(benzo[*c*][1,2,5]thiadiazole-4,7-diyl)bis(2-ethoxybenzohydrazide) (0.03 mmol, 14.8 mg) in a Pyrex tube (10 mL) was sonicated for 3 minutes and followed by addition of acetic acid (6 M, 0.1 mL). The mixture was then degassed via three freeze-pump-thaw cycles. The tube was sealed and heated at 150 °C for 3 days. The precipitate was collected by suction filtration, washed with THF three times, and then Soxhlet extracted by THF overnight. The powder was dried under vacuum at 120 °C to afford Hz-TP-BT-COF with yellow colour in an isolated yield of 79%.

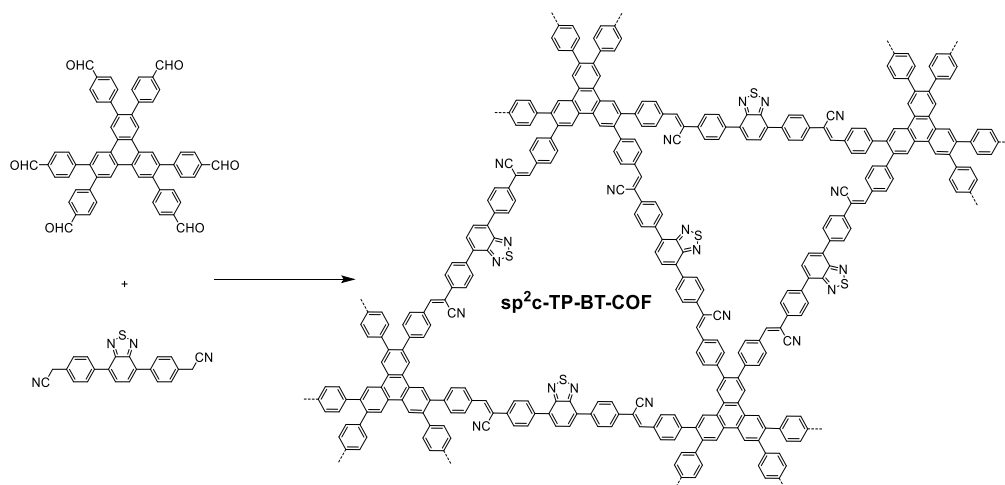
Synthesis of Im-TP-BT-COF



Im-TP-BT-COF: An 1,2-dichlorobenzene/*n*-butanol (*o*-DCB/*n*-BuOH = 0.5 mL/0.5 mL) mixture of 2,3,6,7,10,11-hexakis(4-formylphenyl) triphenylene (0.02 mmol, 17 mg) and 4,4'-(benzo[*c*][1,2,5]thiadiazole-4,7-diyl)dianiline (0.06 mmol, 19.2 mg) in a Pyrex tube (10 mL) was sonicated for 3 minutes and followed by addition of acetic acid (6 M, 0.1 mL). The mixture was then degassed via three freeze-pump-thaw cycles.

The tube was sealed and heated at 120 °C for 3 days. The precipitate was collected by suction filtration, washed with THF three times, and then Soxhlet extracted by THF overnight. The powder was dried under vacuum at 120 °C to afford Im-TP-BT-COF with orange colour in an isolated yield of 91%. Elemental analysis for Im-TP-BT-COF: Calculated: C, 85.65; H, 4.84; N, 6.05; S, 3.46%. Found C, 76.91; H, 4.18; N, 8.30; S, 4.99%

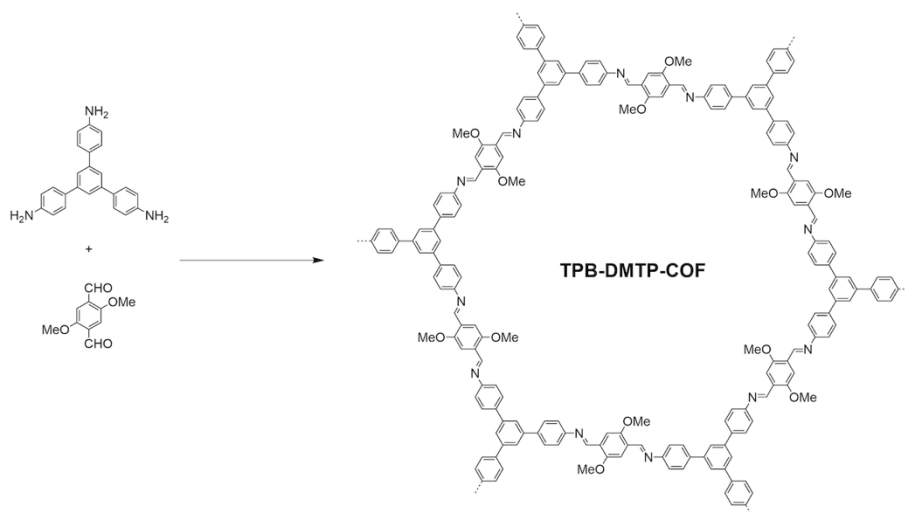
Synthesis of *sp*²*c*-TP-BT-COF



***sp*²*c*-TP-BT-COF:** An *o*-DCB (1 mL) mixture of 2,3,6,7,10,11-hexakis(4-formylphenyl) triphenylene (0.01 mmol, 8.5 mg) and 4,4'-(benzo[*c*][1,2,5]thiadiazole-4,7-diyl)bis(2-ethoxybenzohydrazide) (0.03 mmol, 11.0 mg) in a Pyrex tube (10 mL) was sonicated for 3 minutes and followed by addition of 1,8-diazabicyclo(5.4.0)undec-7-ene (DBU = 0.1 mL). The mixture was then degassed via three freeze-pump-thaw cycles. The tube was sealed and heated at 120 °C for 3 days. The precipitate was collected by suction filtration, washed with THF three times, and then Soxhlet extracted by THF overnight. The powder was dried under vacuum at 120 °C to afford *sp*²*c*-TP-BT with orange-yellow colour in an isolated yield of 85%. Elemental

analysis for sp²c-TP-BT-COF: Calculated: C, 86.36; H, 4.60; N, 5.75; S, 3.29%. Found C, 75.08; H, 4.44; N, 7.39; S, 3.58%.

Synthesis of TPB-DMTP-COF¹



TPB-DMTP-COF: An 1,2-dichlorobenzene/*n*-butanol (*o*-DCB/*n*-BuOH = 0.5 mL/0.5 mL) mixture of 1,3,5-tri(4-aminophenyl)benzene (0.08 mmol, 28.1 mg) and 2,5-dimethoxyterephthalaldehyde (0.12 mmol, 23.4 mg) in a Pyrex tube (10 mL) was sonicated for 3 minutes and followed by addition of acetic acid (6 M, 0.1 mL). The mixture was then degassed via three freeze-pump-thaw cycles. The tube was sealed and heated at 120 °C for 3 days. The precipitate was collected by suction filtration, washed with THF three times, and then Soxhlet extracted by THF overnight. The powder was dried under vacuum at 120 °C to afford TPB-DMTP with orange colour in an isolated yield of 91%.

Supplementary Notes

Supplementary Note 1. Structure simulation, Pawley refinement and atomistic coordinates

Structural geometry optimisation was carried out using BIOVIA Materials Studio suite of programs, using the DFTB module. DFTB+ is an approximate density functional theory method based on the empirical tight-binding approach, utilizing an optimized minimal LCAO Slater-type all-valence basis set in combination with a two-centre approximation for Hamiltonian matrix elements. The lattice dimensions were optimized simultaneously with the geometry. Standard DFTB parameters for X–Y element pair (X, Y = C, H, N, O, S, P) interactions were employed from the *mio* set. Subsequently, Pawley refinement and simulation of the eclipsed AA-stacking mode were predicted using the Reflex module. The lattice parameters were optimised iteratively until the R_{wp} and R_p values converge. The Thompson-Cox-Hastings profile function was used for profile fitting and asymmetry correction function as Berar-Baldinozzi were used during the refinement processes. The atomistic coordinate information was then extracted from the refined structures.

The structures of Hz-TP-BT-COF, Im-TP-BT-COF and sp^2c -TP-BT-COF were also investigated theoretically using fully periodic bulk structural models at the density functional based tight binding (DFTB)² level of theory in AMS program suite³. Based on the experimental data, we have investigated possible AA stacking pattern for all three materials. We have used structural model containing one layer of COF per unit cell, which allows us to fully describe both AA stacking pattern in inclined configuration.

Supplementary Note 2. Electrostatic potential distribution calculations

Electrostatic potential distribution was carried out with geometrically optimised COF structures, using DMol3 module, at GGA-PBE level. The k -point set was at $1 \times 1 \times 4$, and all electrons were considered in core treatment, using the DNP basis set.

Supplementary Note 3. Optoelectronic properties calculations

In order to investigate the opto-electronic properties of Hz-TP-BT-COF, Im-TP-BT-COF and sp^2c -TP-BT-COF, we performed density functional theory (DFT) calculations to determine the band structures of these COFs using Vienna ab initio simulation package (VASP 5.4)⁴. The most stable one layer per unit cell model structures obtained in the DFTB level screening were used and re-optimised at the DFT level.

Supplementary Note 4. Charge analysis calculations

Charge analysis was carried out with geometrically optimised COF structures, using the DMol3 module, at GGA-PBE level. The k -point set was at $1 \times 1 \times 4$, and all electrons were considered in core treatment, using the DNP basis set. The average Mulliken charges were then calculated for each of the structures.

Supplementary Note 5. Photoelectrochemical measurements

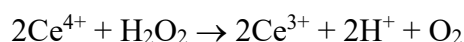
To prepare the COF-coated FTO, as-synthesised COF (2 mg) was added to *N,N*-dimethylformamide (0.5 mL) and 5 wt% Nafion (0.1 mL) and sonicated for 60 min. Then the catalyst ink was spread onto the surface of an FTO plate (1 cm × 1 cm) and dried in a 70 °C oven for 60 min.

Supplementary Note 6. Batch reactor H₂O₂ production

To a 50 mL Schlenk tube was added photocatalyst (3 mg) and deionised water (15 mL). The suspension was well-dispersed by sonication for 5 minutes and bubbled with air for 15 minutes in the dark. A 300W Xenon lamp (Asahi MAX-350 with Mirror module UV-VIS 300 – 600 nm) with 420 nm cut-off filter (to 600 nm) was used to irradiate the reaction mixture with constant coil fan cooling (Supplementary Fig. 16). All apparatus in contact with the reaction (e.g. glass pipette, Schlenk tube) were replaced with new ones or washed thoroughly with deionised water and acetone, and dried completely in 120 °C oven between successive experiments to eliminate cross-contaminations.

Periodically, a portion of the reaction mixture was withdrawn using a syringe and passed through 0.22 µm PTFE filter to remove the photocatalyst. The filtrate (0.1 mL) was then mixed with 0.5 mM Ce(SO₄)₂ in 0.25 M H₂SO₄ (0.9 mL). The concentration of H₂O₂ was then determined from UV-vis absorbance at 316 nm.

The colorimetric analysis of H₂O₂ using Ce(SO₄)₂ is an well-established method⁵, based on the redox reaction, where the yellow Ce⁴⁺ is reduced to colourless Ce³⁺:



By adding known concentrations of H₂O₂ (0.1 mL) to standard solutions of Ce⁴⁺ (0.9 mL), and measuring the absorbance at 316 nm, a calibration curve can be plotted which allows unknown concentrations of H₂O₂ to be interpolated.

Supplementary Note 7. Apparent quantum efficiency measurement

The apparent quantum efficiency (AQE) was determined under monochromatic irradiation using different monochromatic LED lights (420 nm, 450 nm, 540 nm and 620 nm). The measurement was performed in deionised water (25 mL) containing 25 mg of Hz-TP-BT-COF. The light intensity was measured by a ThorLabs PM100D digital optical power and energy meter equipped with S121C sensor. The average light intensity was maintained at ~ 8 W m⁻².

The AQE was calculated using Supplementary equation 1 below:

$$\begin{aligned} \text{AQE (\%)} &= \frac{[\text{H}_2\text{O}_2 \text{ produced (mol)}] \times 2}{\text{photons entered the reactor (mol)}} \times 100 \\ &= \frac{[N_a \times h \times c] [\text{H}_2\text{O}_2 \text{ produced (mol)}] \times 2}{I \times S \times t \times \lambda} \times 100 \quad (1) \end{aligned}$$

where N_a is the Avogadro's constant, h is the Planck constant, c is the speed of light, S is the irradiation area, I is the intensity of light irradiation, t is the irradiation time and λ is the irradiation wavelength.

Supplementary Note 8. Natural sunlight photocatalytic H₂O₂ generation

Natural sunlight photocatalytic H₂O₂ generation using Hz-TP-BT-COF was conducted on a partially cloudy day, from 12:00 to 13:00 hours, on 29 November 2022, at National University of Singapore, Frontier Canteen second level (coordinates: 1.2964847963831054, 103.78068333860926). The Hz-TP-BT-COF (5 mg) was dispersed in air-saturated water (50 mL) in a petri dish with diameter of 86 mm and exposed directly to the sun (Supplementary Fig. 25). The light intensity measured was about 85 mW cm⁻². Quantification of H₂O₂ was performed with the standard curve.

Supplementary Note 9. Recycling experiments

The Hz-TP-BT-COF (75 mg) was dispersed in air-saturated water (100 mL) and irradiated with visible light ($\lambda > 420$ nm) for one hour. After completion of the reaction, the suspension was filtered and the filtrate containing H₂O₂ was analysed using the method described earlier. Hz-TP-BT-COF was then washed with water (100 mL), transferred to a round bottom flask, and dried under vacuum at 120 °C for 2 hours. Hz-TP-BT-COF was then re-weighed before it was used for the next cycle.

Supplementary Note 10. Flow reactor H₂O₂ production

The column was homemade. In detail, a glass rod (external diameter: 5 mm) was put in the middle of a condenser (Internal diameter: 7 mm) and then 90 mg of Hz-TP-BT-COF was packed in the free space between the condenser and glass rod. The effective column length is 72 mm with a resistance time of 80 seconds.

The flow reactor system (Supplementary Fig. 28) consists of air-saturated water, flow pump (Ou Shisheng DP-S10), Hz-TP-BT-COF column, LED light and H₂O₂ collection container, connected with PTFE tubing (I.D. = 1 mm, O.D. = 1.6 mm). The air-saturated H₂O was pumped at a rate of 0.75 mL min⁻¹ into the column, where two blue LEDs ($\lambda = 460$ nm, 30 W) were set at a distance of 3 cm. The generated H₂O₂ was received with a collection container and stored in the dark. The total volume of H₂O₂ solution collected over 144 hours was 6.48 litres, stored in five-litre bottles (Supplementary Fig. 29).

Supplementary Note 11. Gas-chromatography – mass spectroscopy study on flow reactor

GC-MS analysis was performed on Shimadzu GC-2010 gas chromatography coupled to a Shimadzu QP2010 mass selective detector. 10 mL of H₂O₂ solution obtained from flow reactor was transferred into a round bottle flask and evaporated to dryness. Then, 2 mL of ethyl acetate was added to rinse the flask. The solvent was collected for GC-MS analysis. Only baseline was observed (Supplementary Fig. 30).

Supplementary Note 12. Gravimetric measurement after continuous flow

After the flow reaction, Hz-TP-BT-COF was removed from the column, washed with water and ethanol, dried under hot vacuum at 120 °C, before subsequent characterisations. The final mass recorded was 88.3 mg, suggesting a weight loss of only 1.89 % (Supplementary Fig. 35).

Supplementary Note 13. Identification of superoxide radical anion using the nitro blue tetrazolium (NBT) method

The generation of superoxide radical anion ($\cdot\text{O}_2^-$) can be measured by the decay of NBT using UV-vis spectroscopy⁶, which has an absorbance maximum at 259 nm (Supplementary Fig. 37).

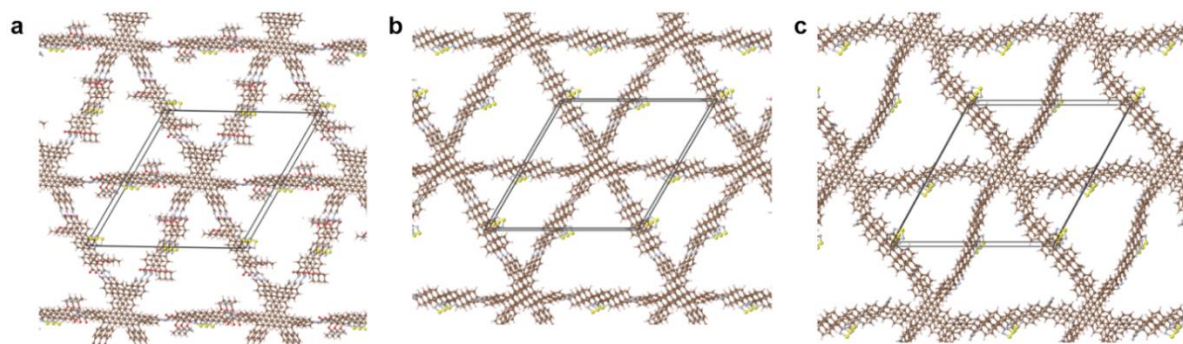
Supplementary Note 14. Gibbs free energy calculations

To simulate the H₂O₂ generation reaction and the OER, we have constructed a 1×1×3 supercell of the COFs (Fig. 6b and c) to better represent the actual scenario, as well as the micro-environment. And we have applied computational hydrogen electrode model to derive the Gibbs free energies with the Nørskov's approach (see Methods, computational details).

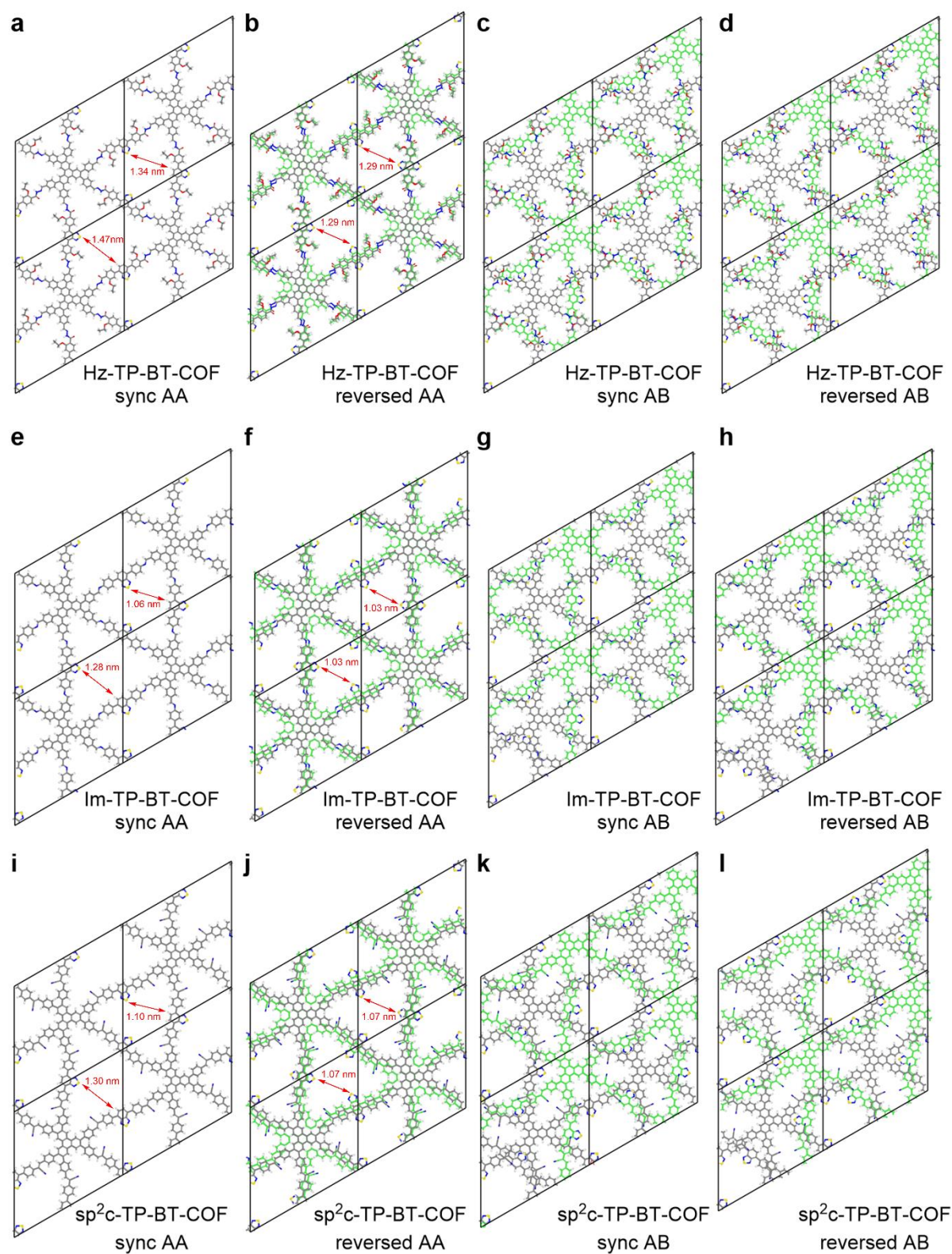
Supplementary Note 15. Isotopic labelling experiment

To a 1.5 mL glass vial was added Hz-TP-BT-COF (2 mg) and H₂¹⁸O (0.5 mL). The suspension was well-dispersed by sonication for 5 minutes and bubbled with air for 15 minutes in the dark, before sealing off with a rubber septum. A 300W Xenon lamp (Asahi MAX-350 with Mirror module UV-VIS 300 – 600 nm) with 420 nm cut-off filter (to 600 nm) was used to irradiate the reaction mixture with constant coil fan cooling for 12 hours before analysis of headspace (H768) by GC-MS. Thereafter, we purged the reaction with argon, extracted the aqueous phase, removed the photocatalyst, before injecting it into a vial containing MnO₂ (oxidises H₂O₂ to O₂). We then performed headspace sampling and GC-MS analysis, which revealed the formation of ¹⁸O₂ (Supplementary Fig. 40 and 41).

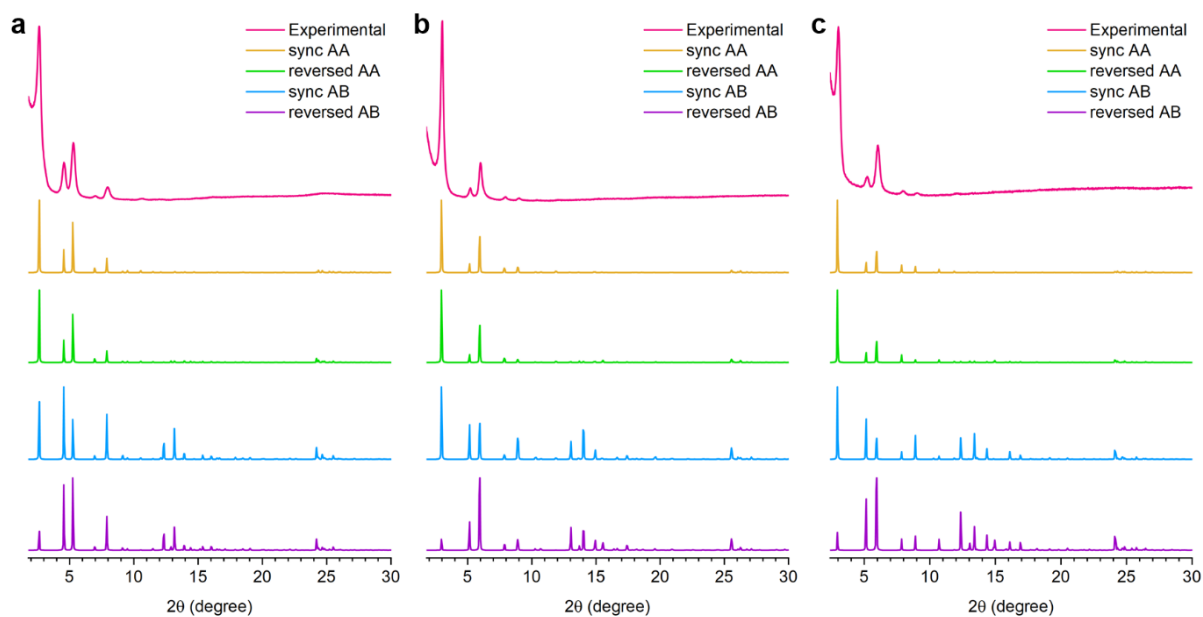
Supplementary Figures



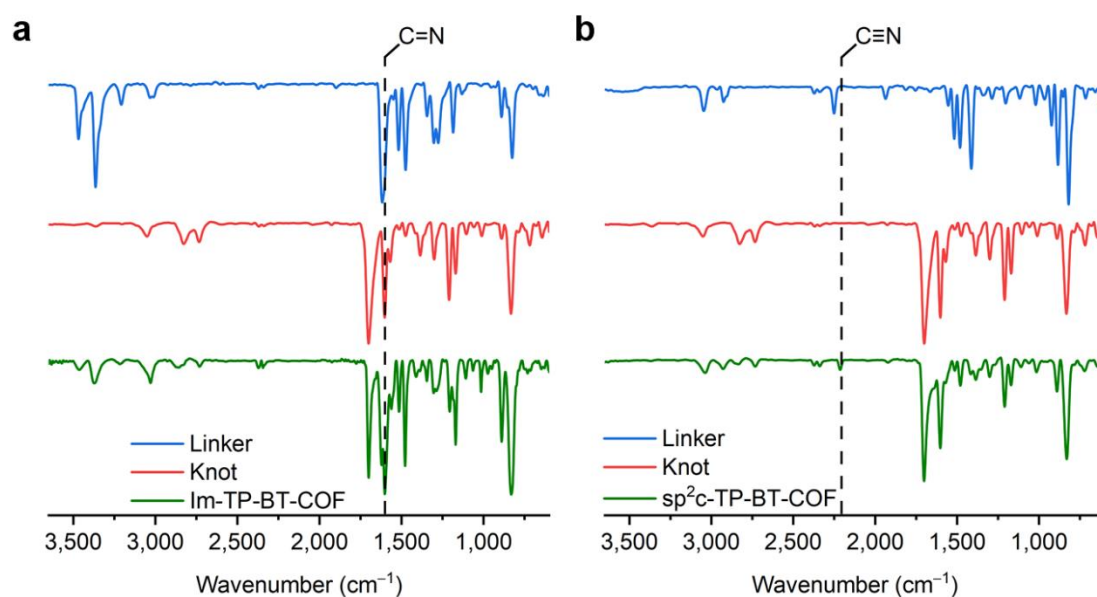
Supplementary Fig. 1 | Optimised structures. a–c, One layer per unit cell structural model at the DFTB level for Hz-TP-BT-COF (a), Im-TP-BT-COF (b) and sp^2c -TP-BT-COF (c).



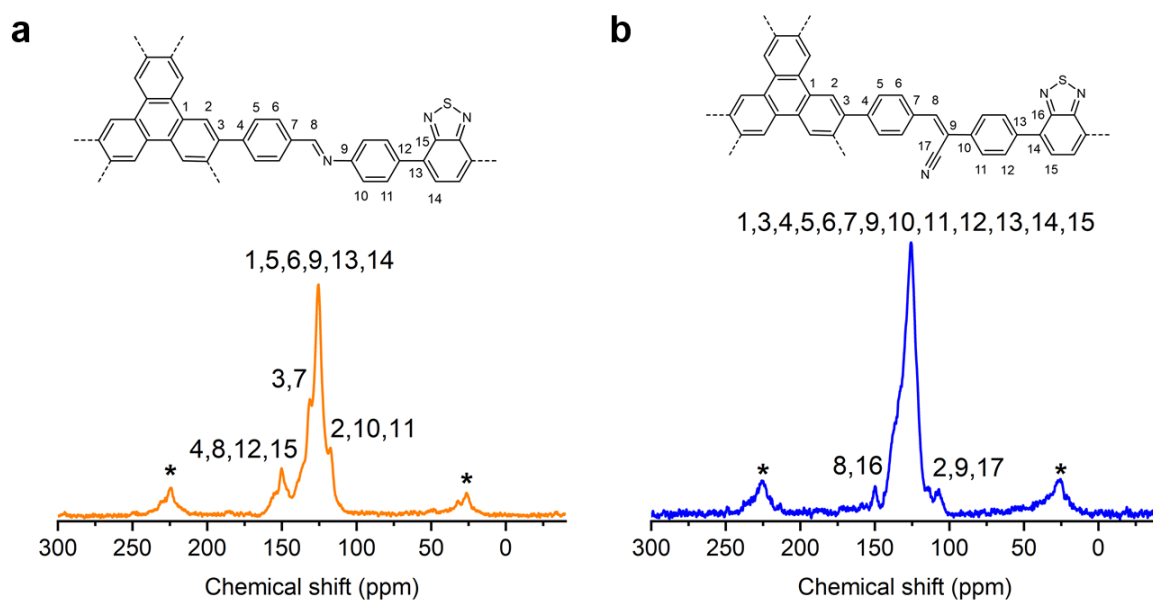
Supplementary Fig. 2 | Optimised stacking modes. a–l, Various stacking modes for Hz-TP-BT-COF (a–d), Im-TP-BT-COF (e–h) and sp²c-TP-BT-COF (i–l).



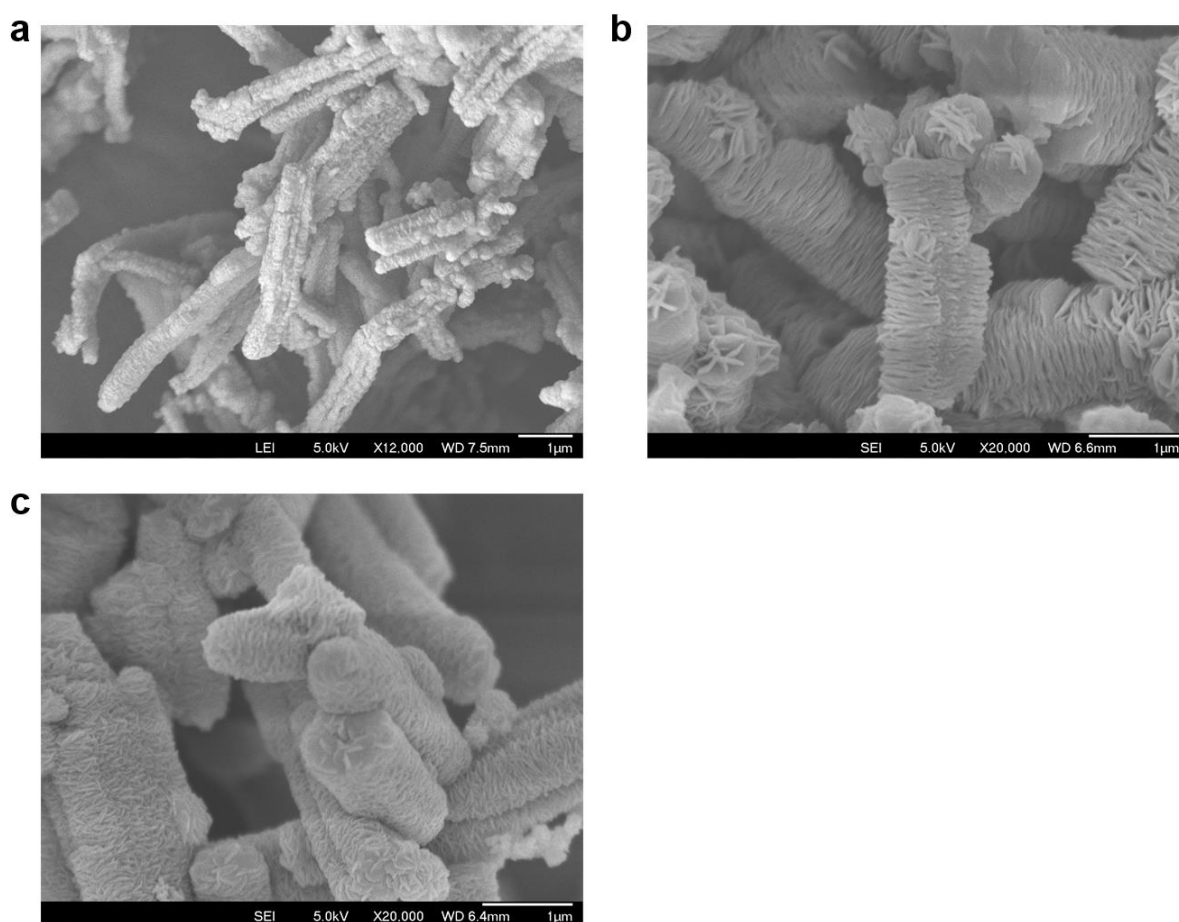
Supplementary Fig. 3 | Simulated PXRD patterns for different stacking modes. a-c, PXRD patterns for different stacking modes of Hz-TP-BT-COF (a), Im-TP-BT-COF (b) and sp²c-TP-BT-COF (c).



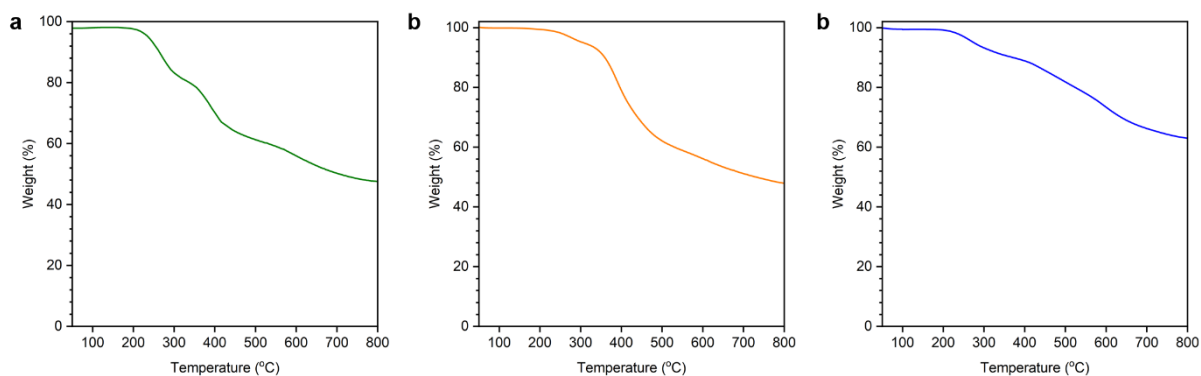
Supplementary Fig. 4 | FT-IR spectrum. a-b, FTIR spectrum of Im-TP-BT-COF (a) and sp²c-TP-BT-COF (b).



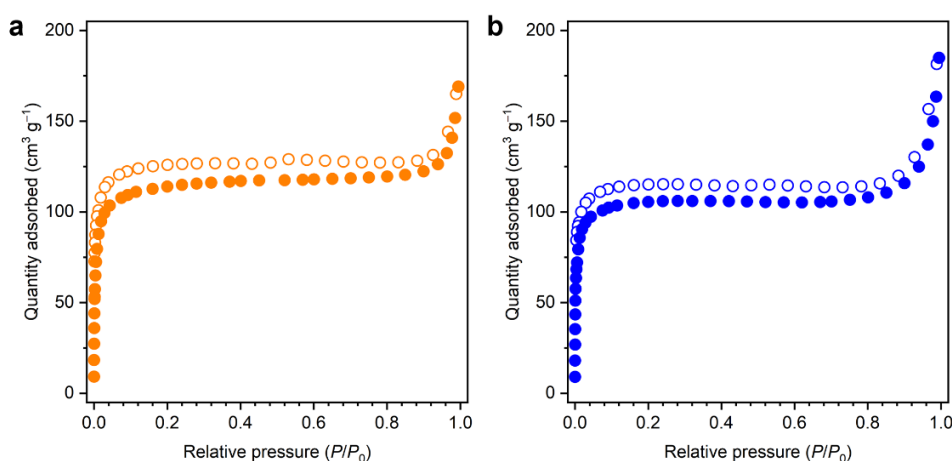
Supplementary Fig. 5 | Solid state ^{13}C CP/MAS NMR spectrum. a-b, Solid state NMR of Im-TP-BT-COF (a) and sp^2c -TP-BT-COF (b).



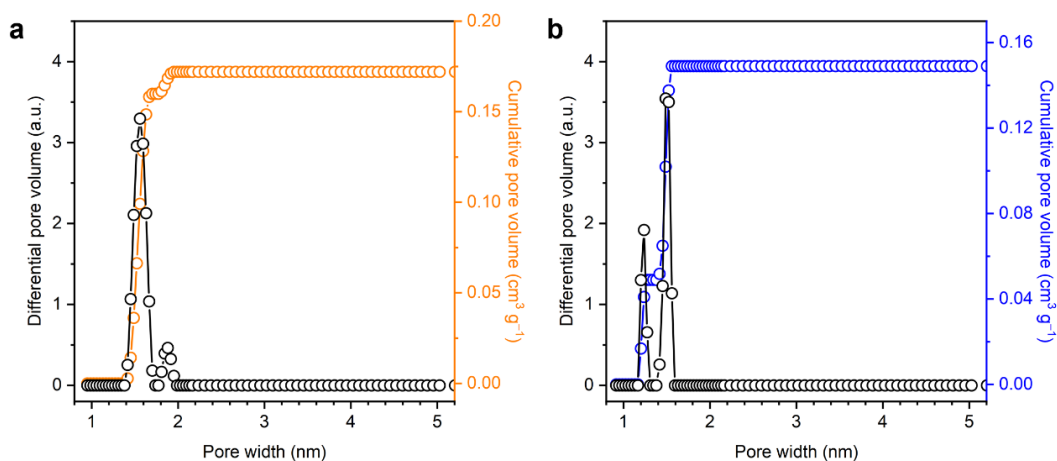
Supplementary Fig. 6 | Field emission scanning electron microscopy images. a-c, FESEM images of Hz-TP-BT-COF (a), Im-TP-BT-COF (b) and sp^2c -TP-BT-COF (c).



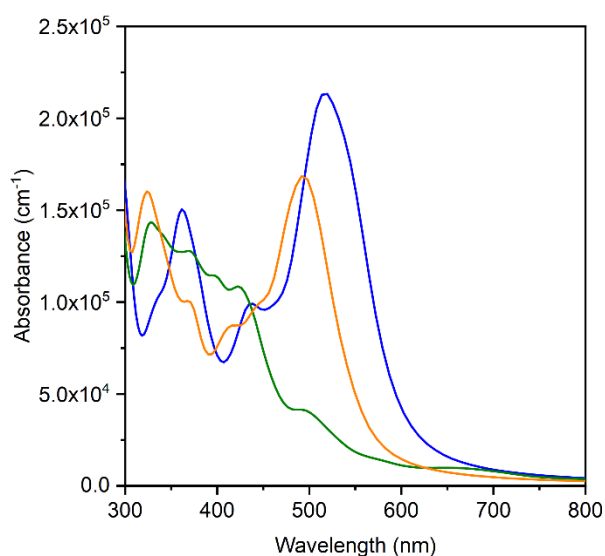
Supplementary Fig. 7 | TGA trace. a-c, TGA traces of Hz-TP-BT-COF (a), Im-TP-BT-COF (b) and sp²c-TP-BT-COF (c) under N₂ atmosphere from 40 – 800 °C.



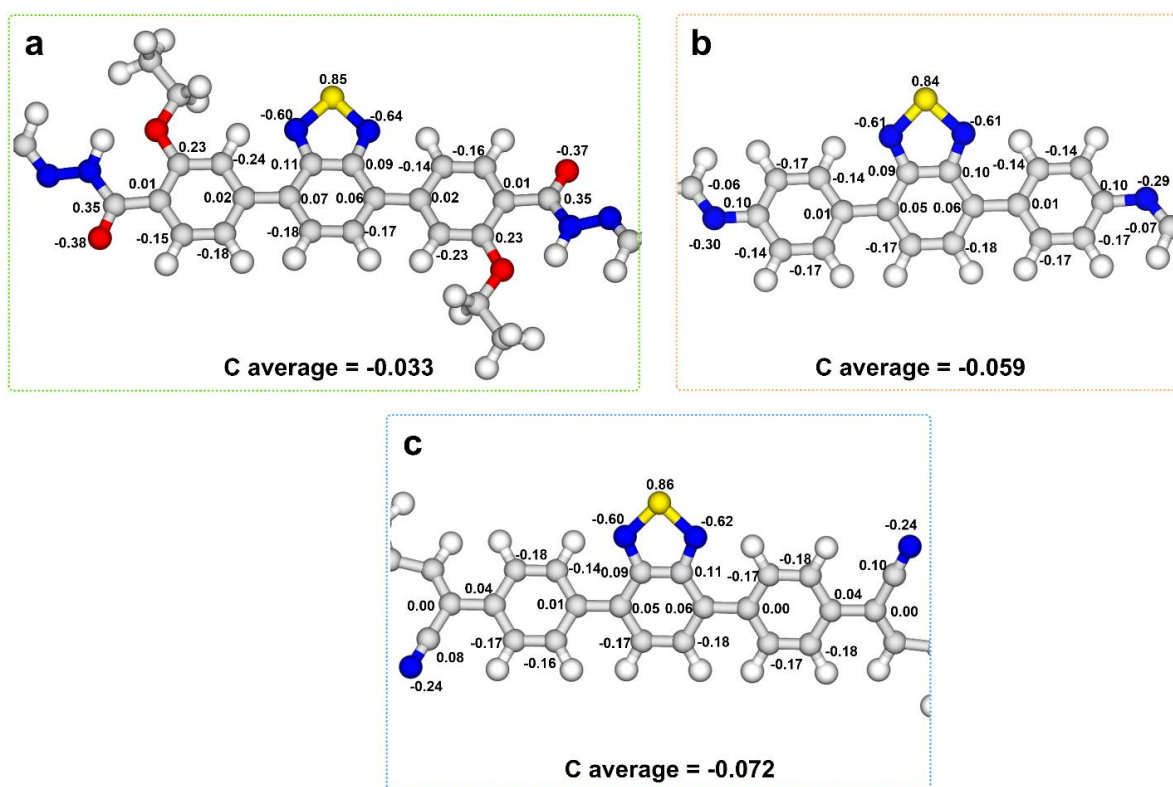
Supplementary Fig. 8 | N₂ sorption profile. a, b, Nitrogen sorption isotherms of Im-TP-BT-COF (a) and sp²c-TP-BT-COF (b).



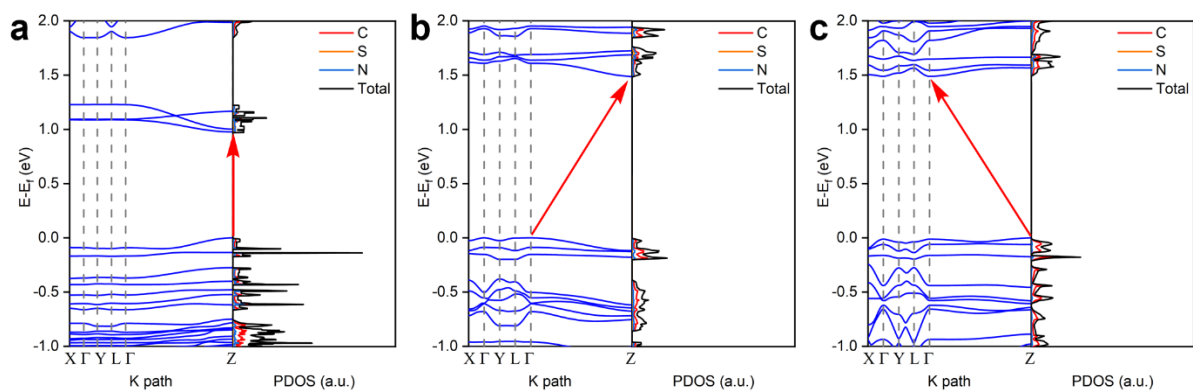
Supplementary Fig. 9 | Porosity. a, b, Cumulative pore volumes and pore size distribution curves for Im-TP-BT-COF (a) and sp²c-TP-BT-COF (b).



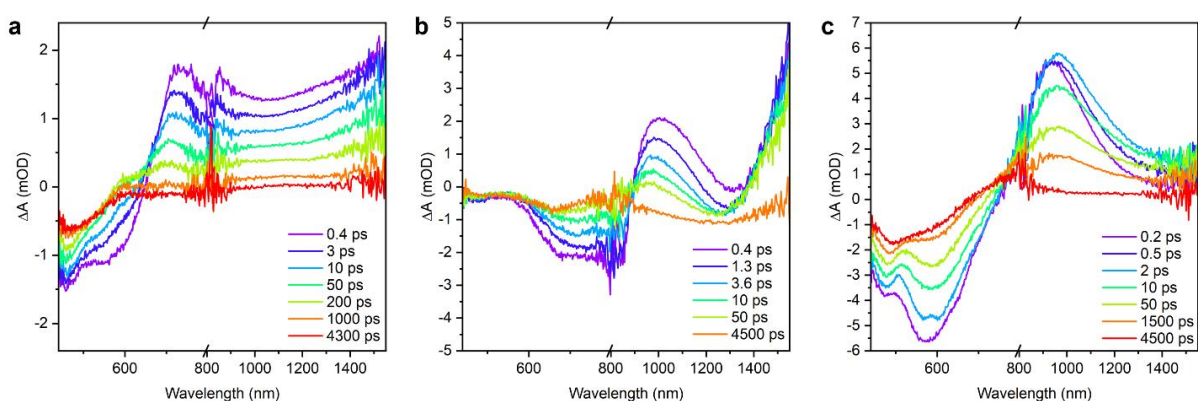
Supplementary Fig. 10 | Calculated electronic absorption spectra. Electronic absorption spectra of Hz-TP-BT-COF (green curve), Im-TP-BT-COF (orange curve) and sp^2c -TP-BT-COF (blue curve).



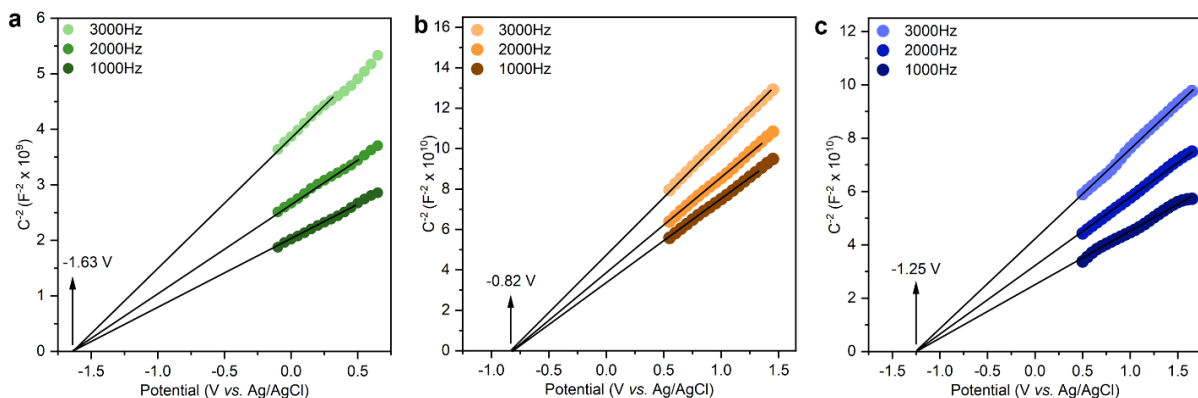
Supplementary Fig. 11 | Charge distribution. a–c, Charge distribution on linker units in Hz-TP-BT-COF (a), Im-TP-BT-COF (b) and sp^2c -TP-BT-COF (c).



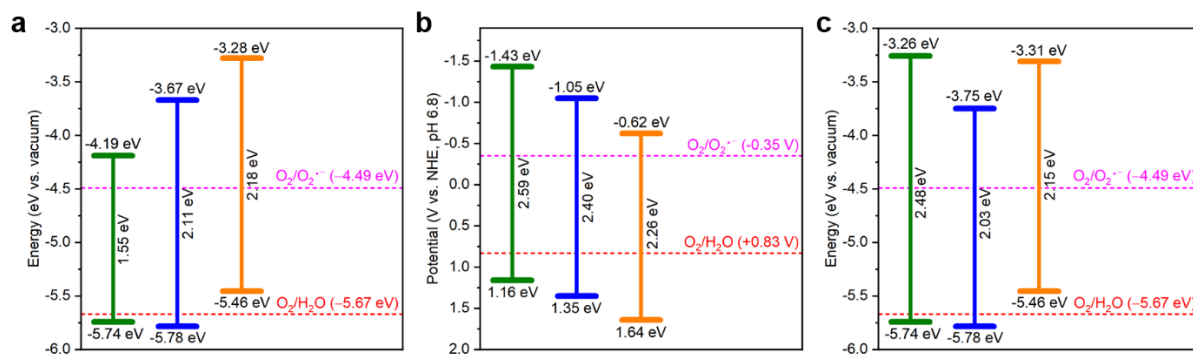
Supplementary Fig. 12 | Band structure and PDOS. a-c, Band structure and PDOS for Hz-TP-BT-COF (a), Im-TP-BT-COF (b) and sp²c-TP-BT-COF (c).



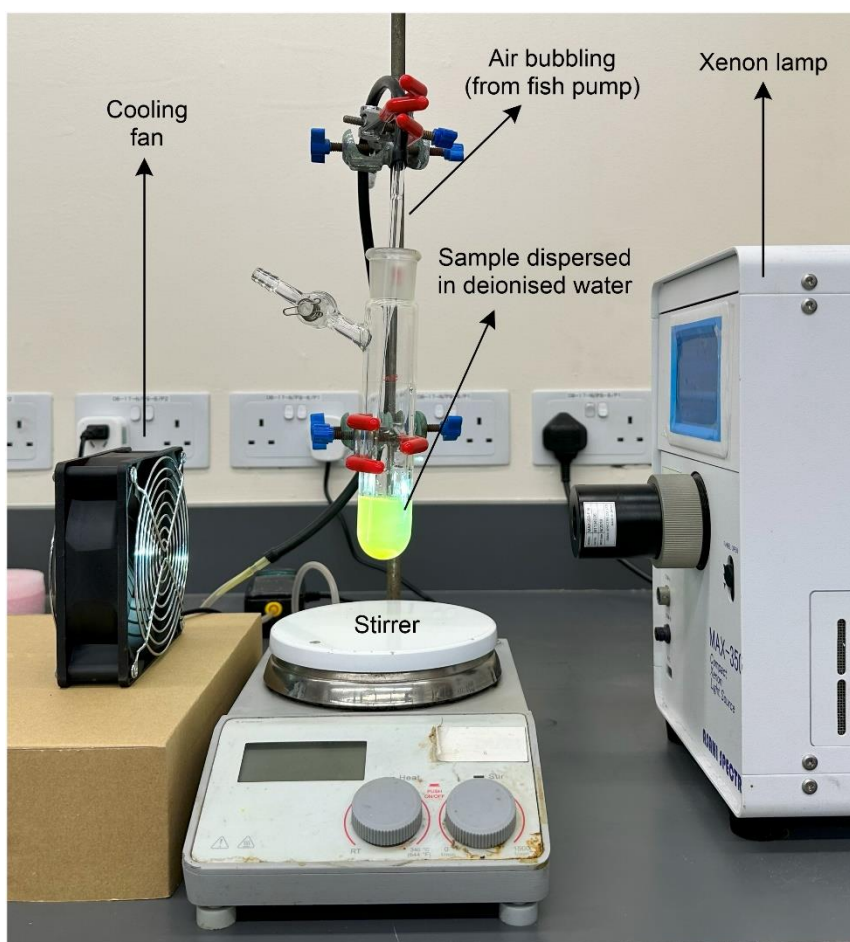
Supplementary Fig. 13 | Transient absorption spectroscopy. a-c, Transient absorption spectra of Hz-TP-BT-COF (a), Im-TP-BT-COF (b) and sp²c-TP-BT-COF (c).



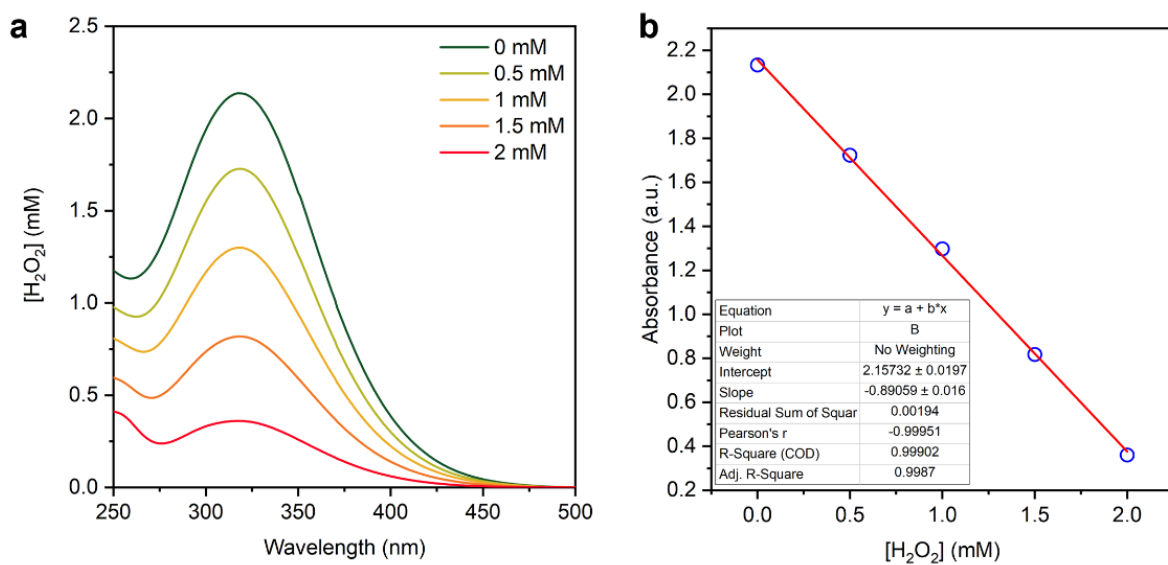
Supplementary Fig. 14 | Photoelectrochemical results. a-c, Mott-Schottky plots for Hz-TP-BT-COF (a), Im-TP-BT-COF (b) and sp²c-TP-BT-COF (c).



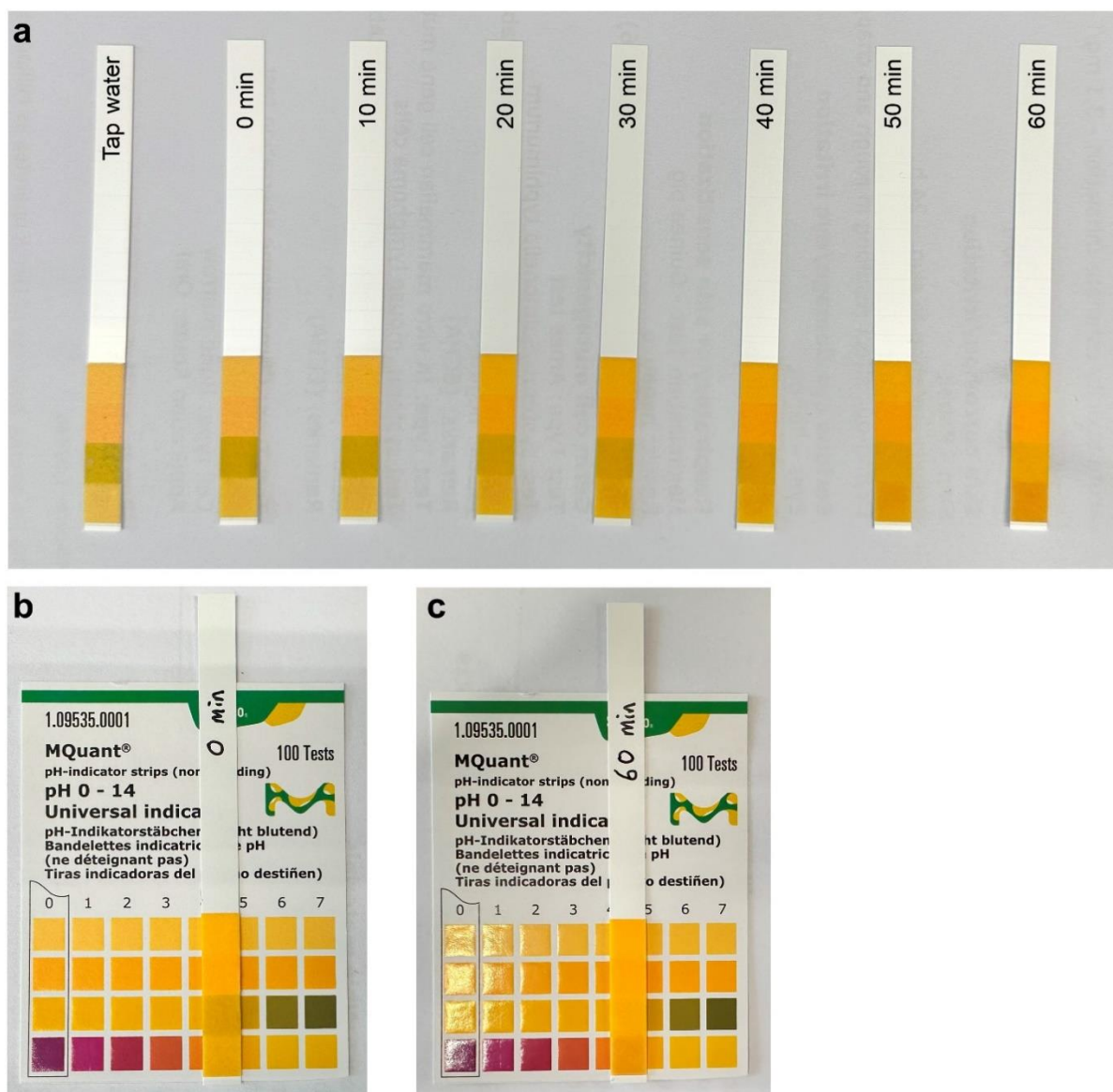
Supplementary Fig. 15 | Energy band alignments of the COFs. a, DFT derived. **b**, Experimentally derived. **c**, TD-DFT derived. Green colour: Hz-TP-BT-COF, blue colour: sp²c-TP-BT-COF, orange colour: Im-TP-BT-COF.



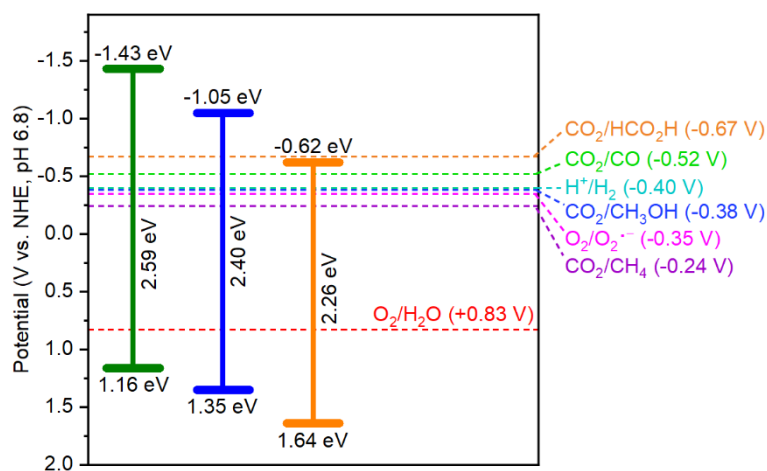
Supplementary Fig. 16 | Digital photograph of photocatalytic reactor setup.



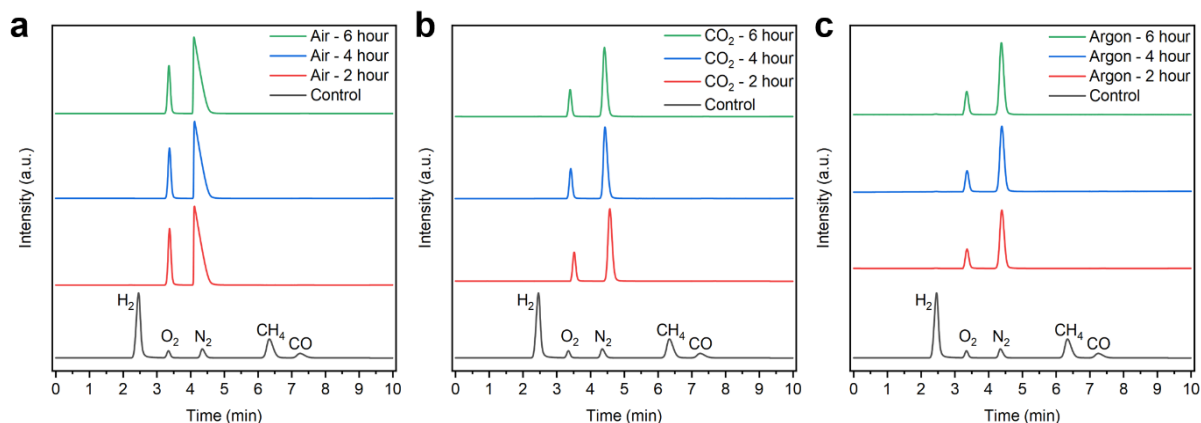
Supplementary Fig. 17 | Calibration curves. **a**, Changes in absorbance of Ce^{4+} with different concentrations of H_2O_2 . **b**, Calibration curve of absorbance at 316 nm against concentration of H_2O_2 .



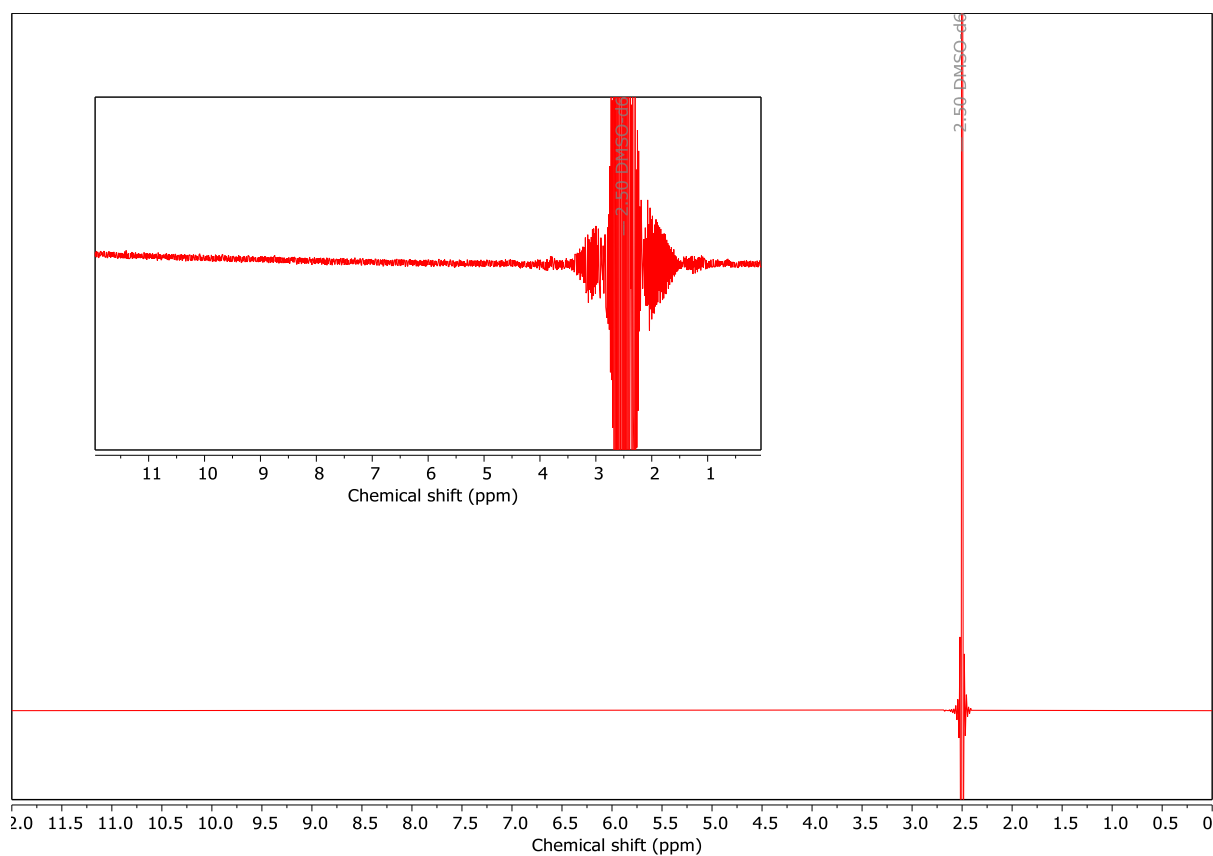
Supplementary Fig. 18 | pH monitoring. **a**, pH changes during course of photocatalytic reaction. **b**, Comparison of pH against pH scale at $t = 0$ min. **c**, Comparison of pH against pH scale at $t = 60$ min.



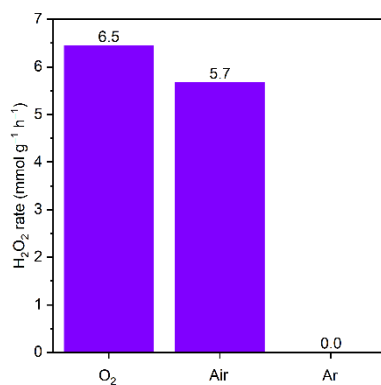
Supplementary Fig. 19 | Energy band alignments of the COFs against other reduction potentials. Green colour: Hz-TP-BT-COF, blue colour: sp²c-TP-BT-COF, orange colour: Im-TP-BT-COF.



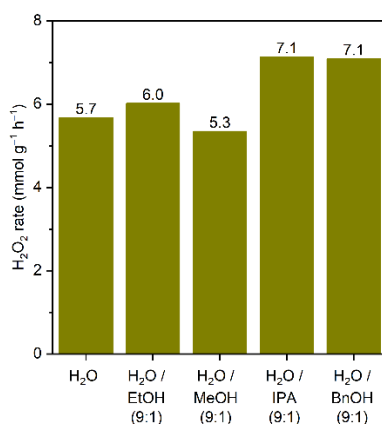
Supplementary Fig. 20 | GC chromatogram (TCD detector) of photocatalytic experiment performed using Hz-TP-BT-COF. a-c, Under air (a), carbon dioxide (b) and argon atmosphere (c).



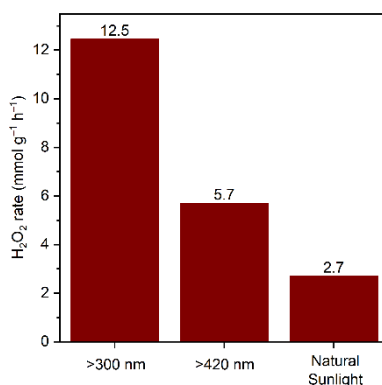
Supplementary Fig. 21 | ^1H NMR with water suppression of aqueous phase of photocatalytic reaction. Inset shows enlarged graph.



Supplementary Fig. 22 | Photocatalytic H_2O_2 generation of Hz-TP-BT-COF in water saturated with different gases, under visible light irradiation ($\lambda > 420$ nm).



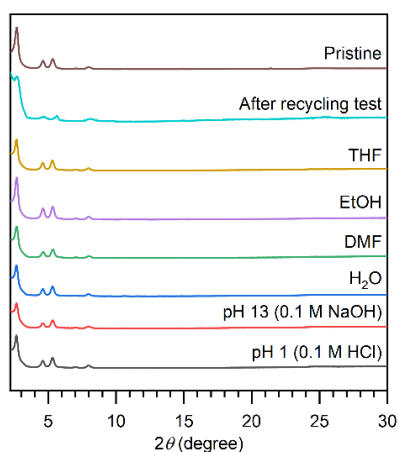
Supplementary Fig. 23 | Photocatalytic H_2O_2 generation of Hz-TP-BT-COF after adding alcohols as sacrificial donors, under visible light irradiation ($\lambda > 420$ nm) in air-saturated aqueous solutions.



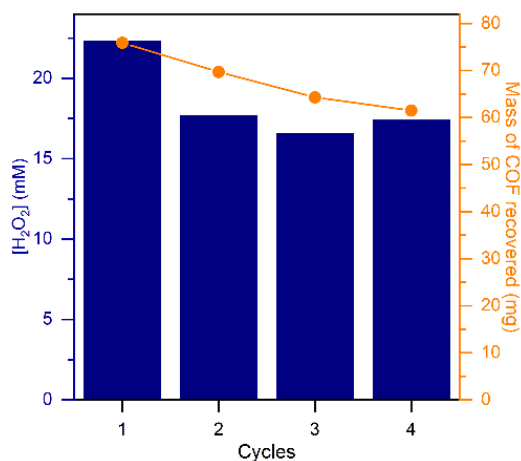
Supplementary Fig. 24 | Photocatalytic H_2O_2 generation of Hz-TP-BT-COF under different light irradiation, in air-saturated water.



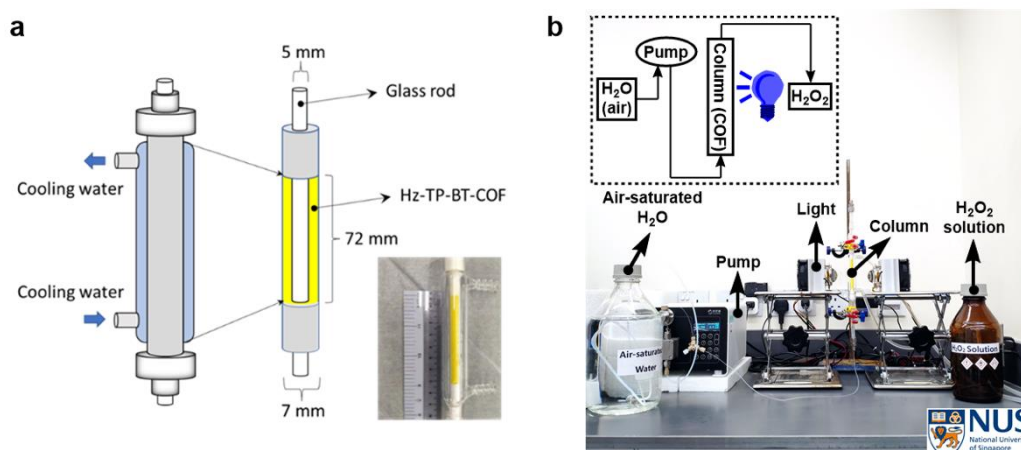
Supplementary Fig. 25 | Digital photograph of the set up for natural sunlight photocatalytic H_2O_2 generation using Hz-TP-BT-COF.



Supplementary Fig. 26 | Stability tests for Hz-TP-BT-COF.



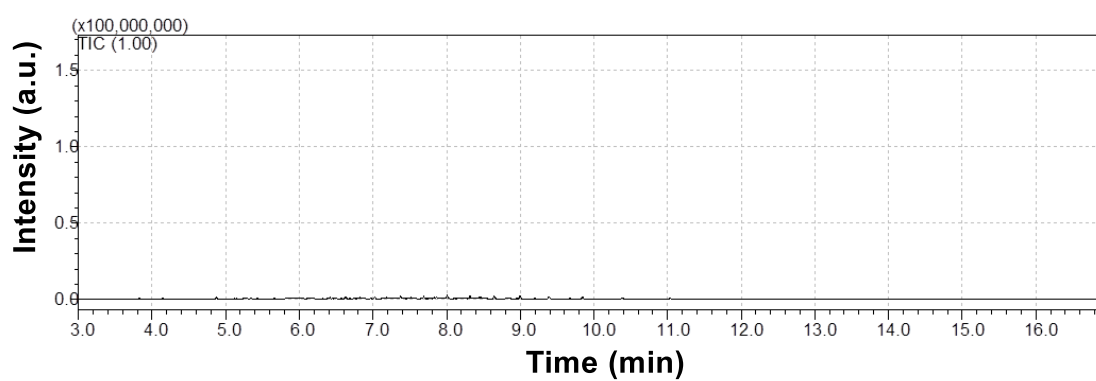
Supplementary Fig. 27 | Tests of Hz-TP-BT-COF over four cycles.



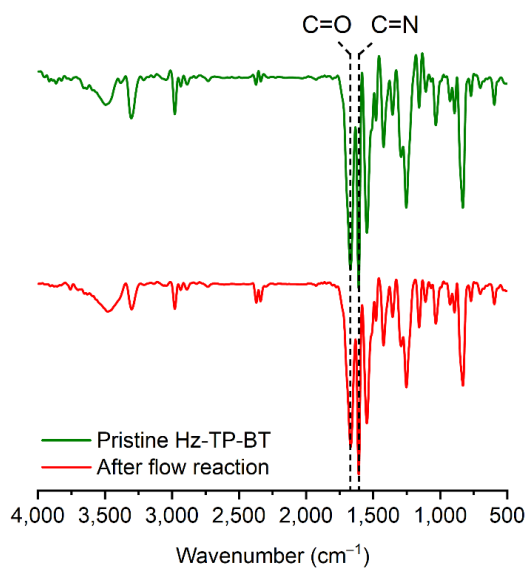
Supplementary Fig. 28 | Flow-type reactor setup. **a**, Schematic of column assembly and digital image of the Hz-TP-BT-COF column. **b**, Setup of flow reactor.



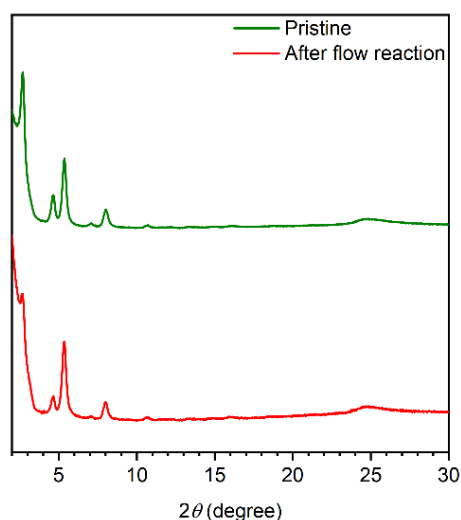
Supplementary Fig. 29 | Digital photographs of produced H₂O₂ solution from the flow reactor.



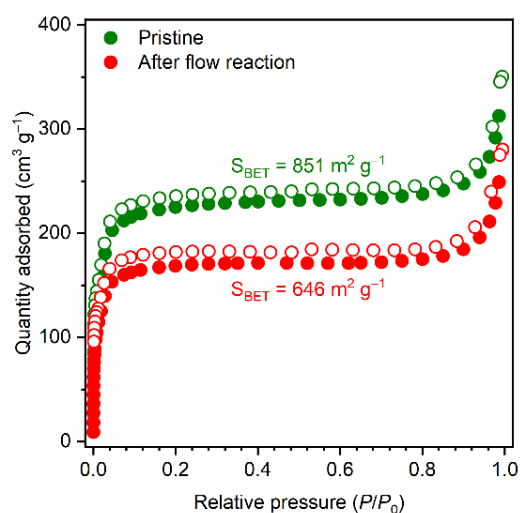
Supplementary Fig. 30 | GC-MS Chromatogram (TIC) of the collected H₂O₂ solution.



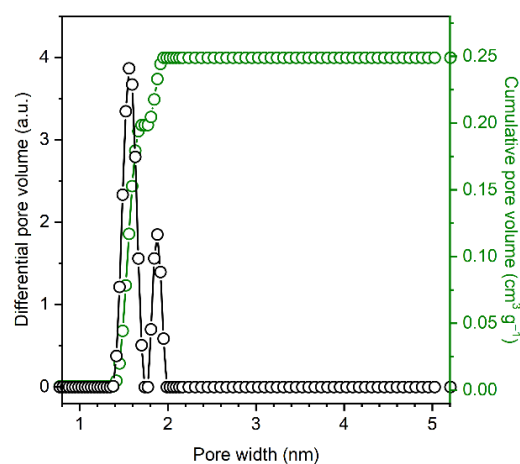
Supplementary Fig. 31 | Comparison of FT-IR spectra of Hz-TP-BT-COF before (green curve) and after flow reaction (red curve).



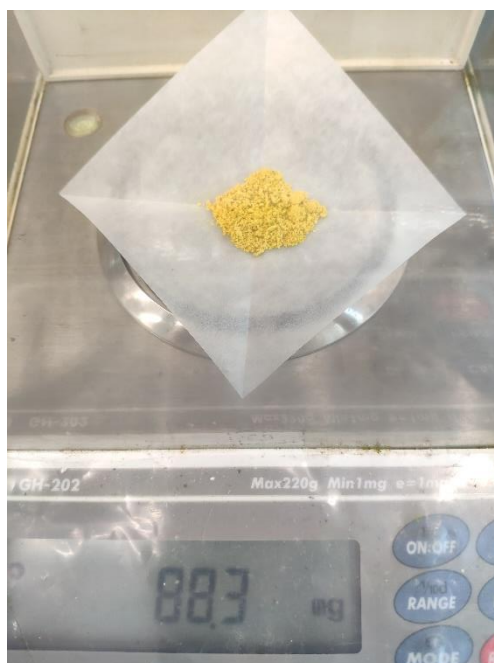
Supplementary Fig. 32 | PXRD patterns of Hz-TP-BT-COF before (green curve) and after (red curve) flow reaction.



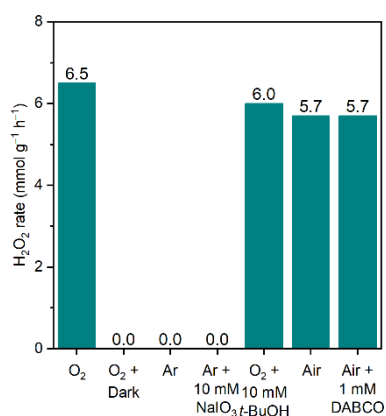
Supplementary Fig. 33 | Nitrogen sorption isotherms of Hz-TP-BT-COF before (green curve) and after (red curve) the flow reactor photocatalytic H_2O_2 generation.



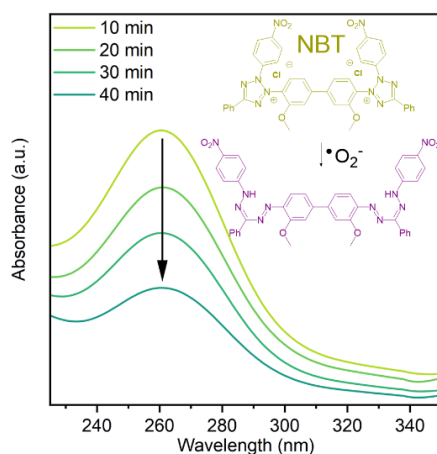
Supplementary Fig. 34 | Cumulative pore volume (green curve) and pore size distribution (black curve) profiles of Hz-TP-BT-COF after the flow reactor photocatalytic H_2O_2 generation.



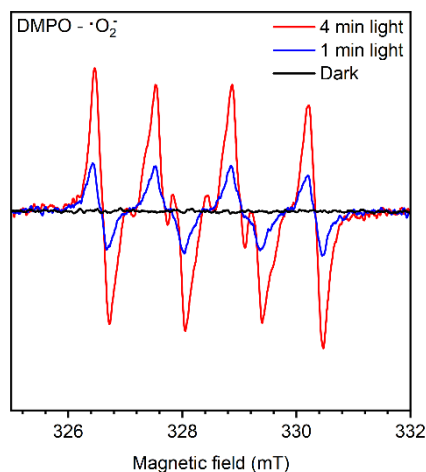
Supplementary Fig. 35 | Digital photograph of recycled Hz-TP-BT-COF and its weight after structural characterisation.



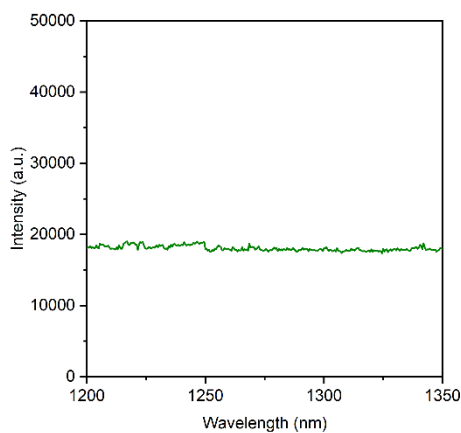
Supplementary Fig. 36 | Photocatalytic H_2O_2 generation of Hz-TP-BT-COF under different reaction conditions.



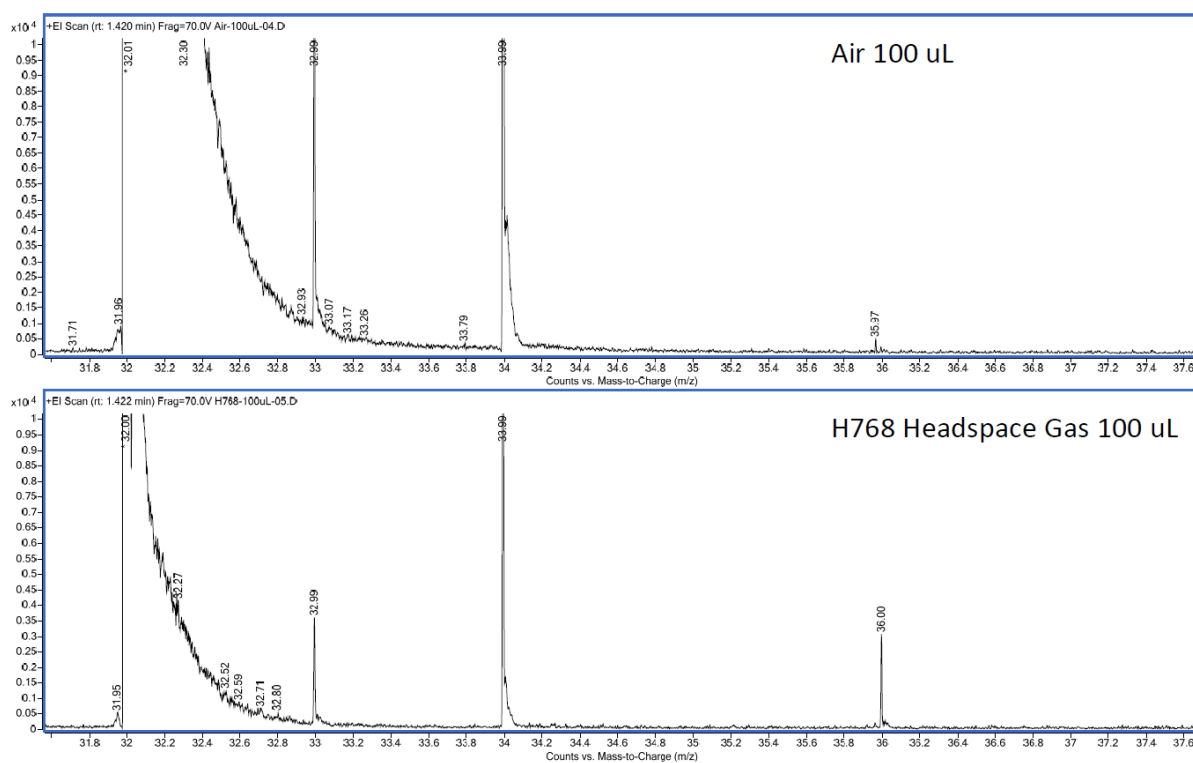
Supplementary Fig. 37 | Decrease in absorbance of NBT due to the formation of superoxide radical anion.



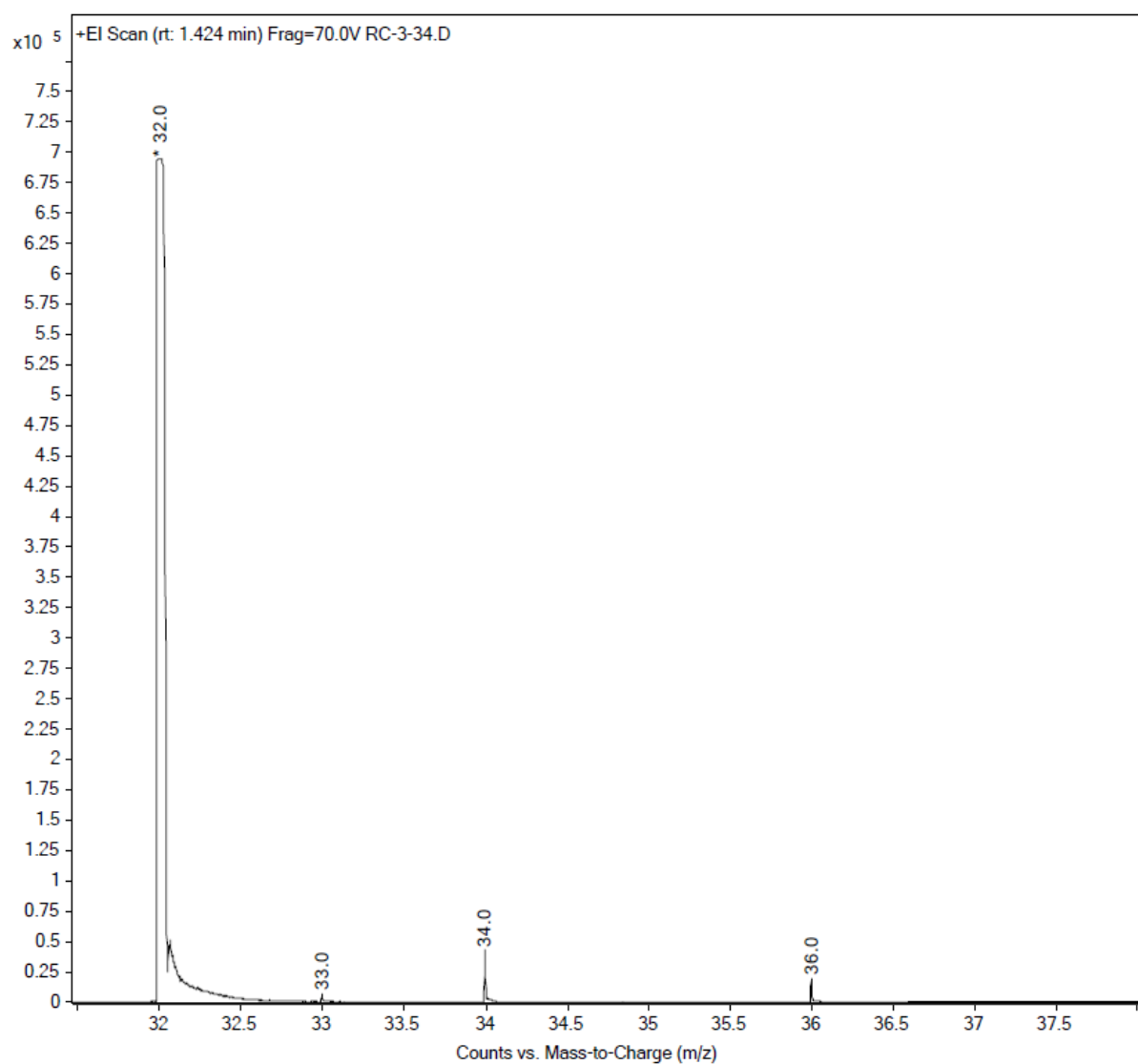
Supplementary Fig. 38 | EPR spectrum of Hz-TP-BT-COF in the dark (black curve) and upon irradiation (blue curve: 1-min irradiation, red curve: 4-min irradiation) with a 30-W blue LED ($\lambda = 460$ nm).



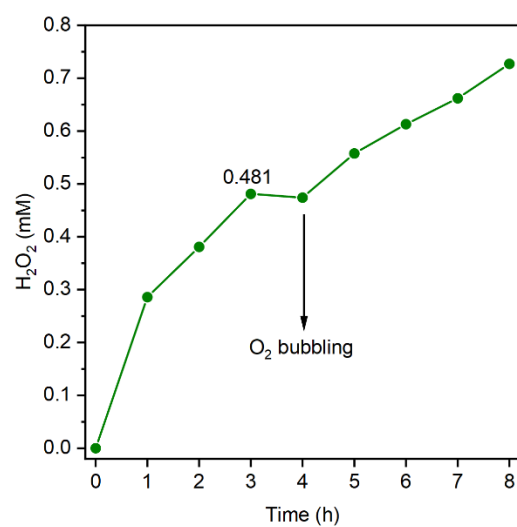
Supplementary Fig. 39 | Phosphorescence measurement did not detect characteristic singlet oxygen phosphorescence at 1270 nm.



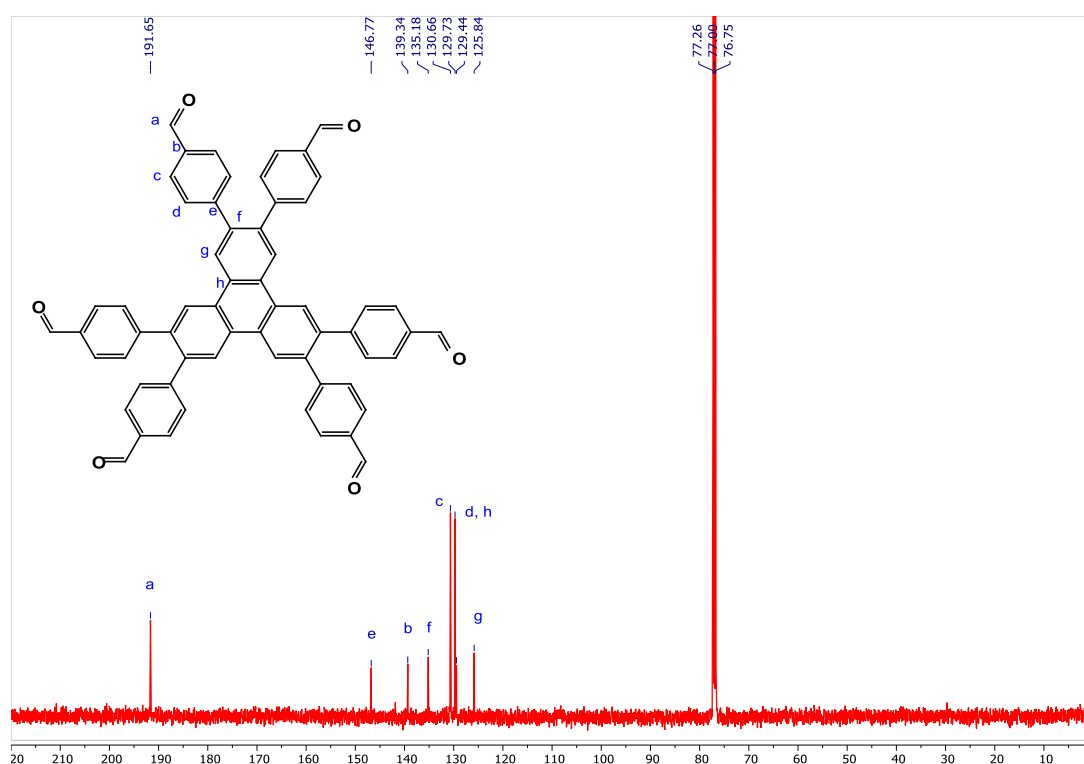
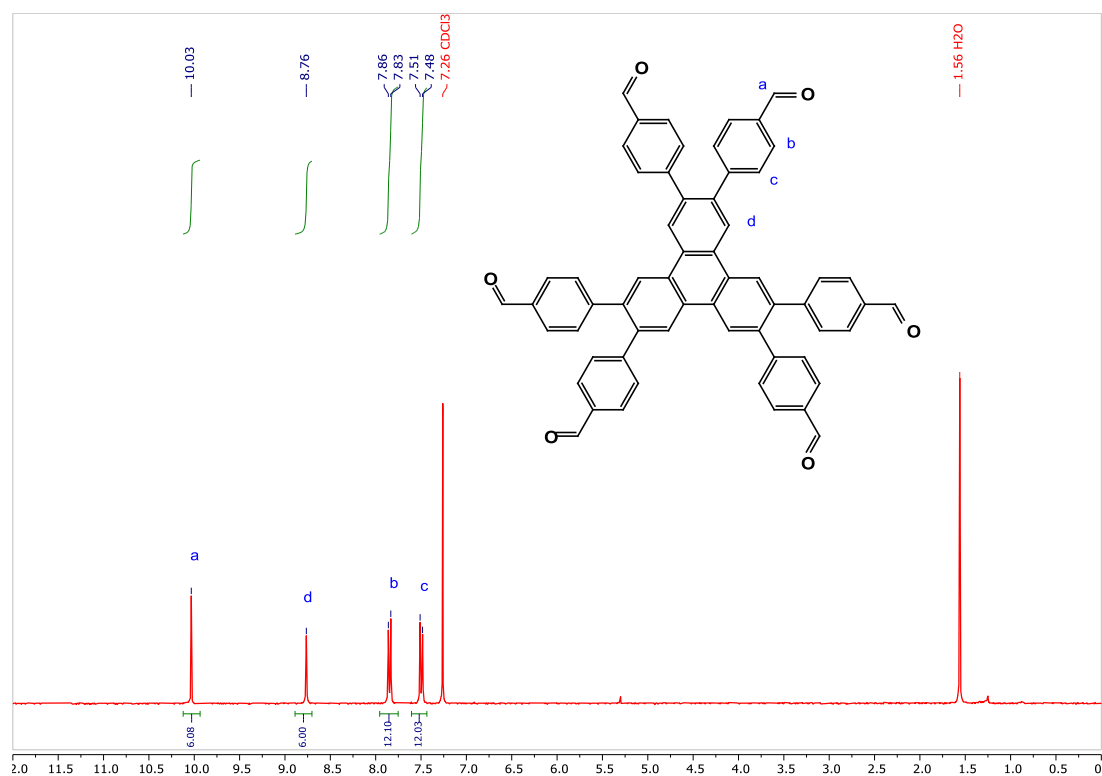
Supplementary Fig. 40 | GC-MS Chromatogram of the headspace. GC-MS Chromatogram of the air (100 μ L; top panel) and of the headspace of reaction system with H_2^{18}O (bottom panel).



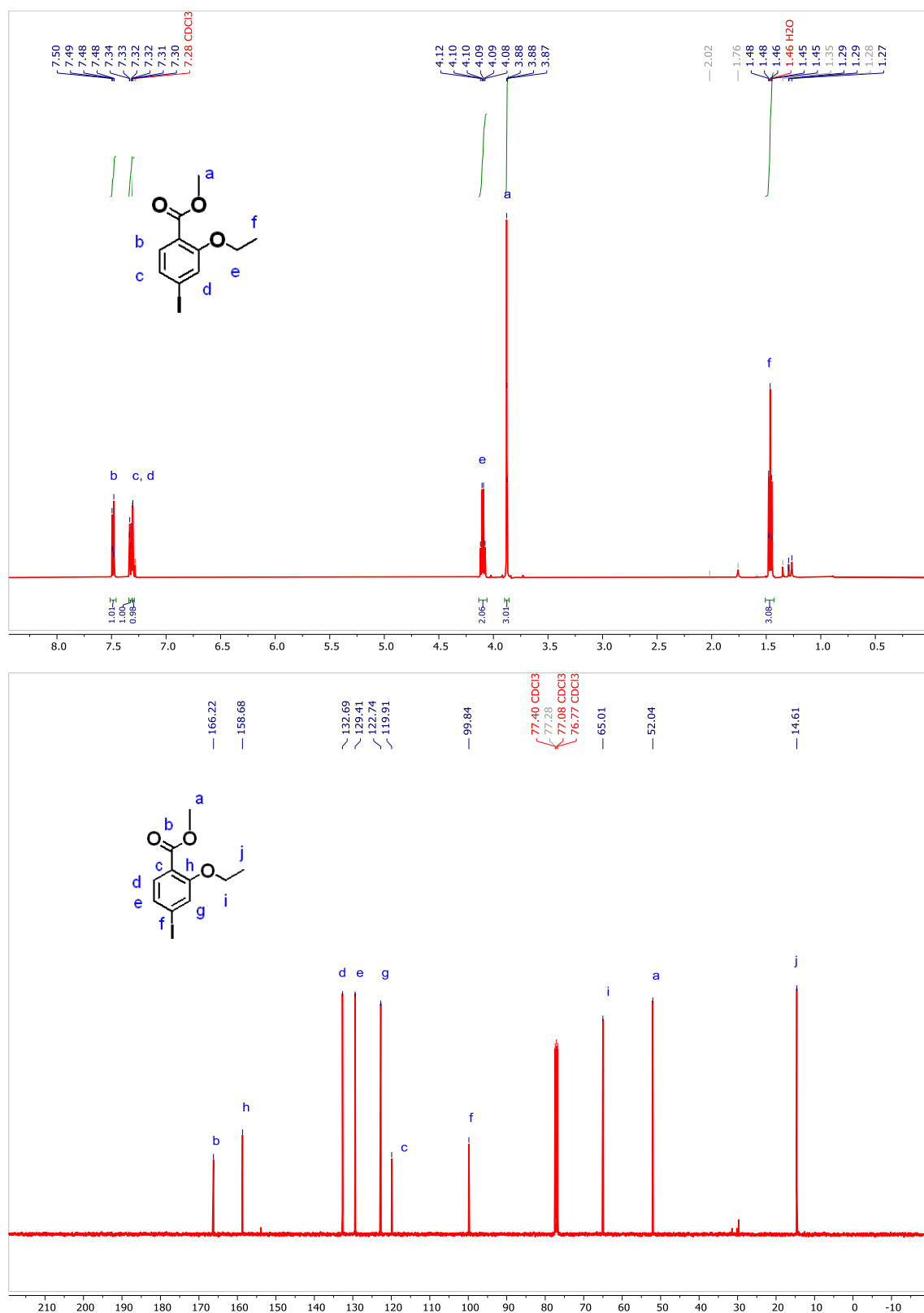
Supplementary Fig. 41 | GC-MS Chromatogram of the headspace after decomposition with MnO₂.



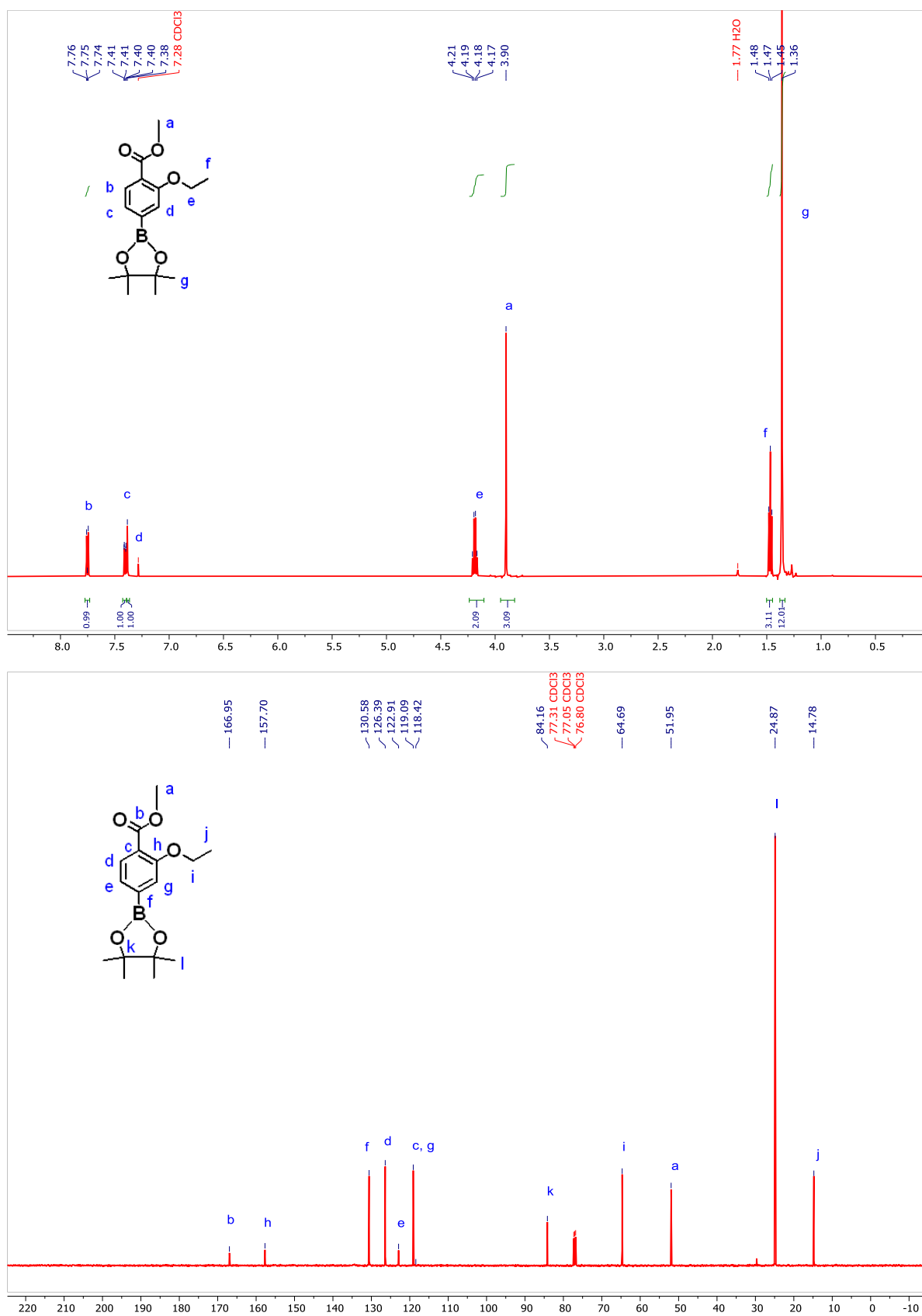
Supplementary Fig. 42 | Photosynthesis of H_2O_2 using Hz-TP-BT-COF under pre- O_2 saturated water in a closed environment.



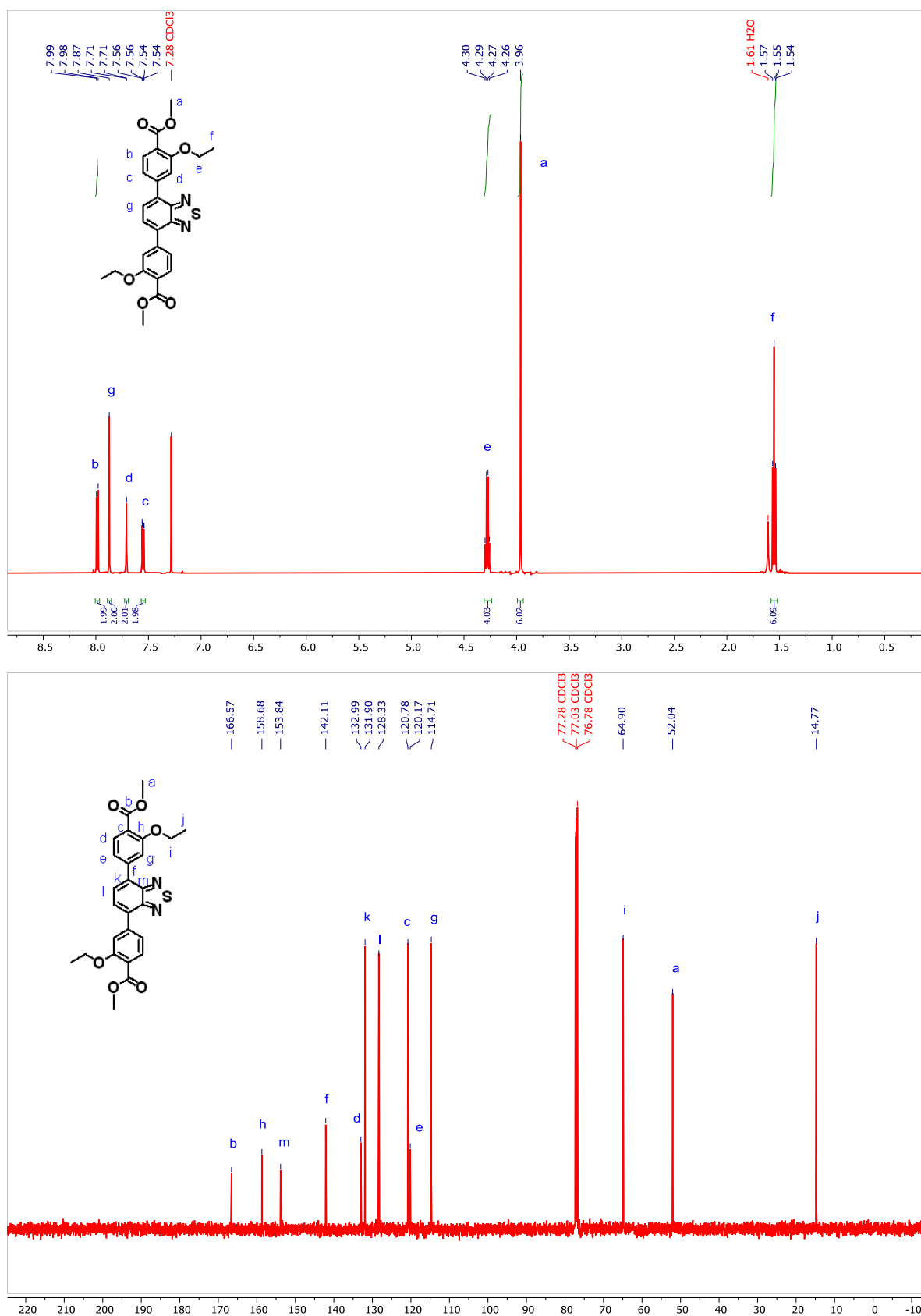
Supplementary Fig. 43 | ¹H NMR and ¹³C NMR spectra of 2,3,6,7,10,11-hexakis(4-formylphenyl) triphenylene (B).



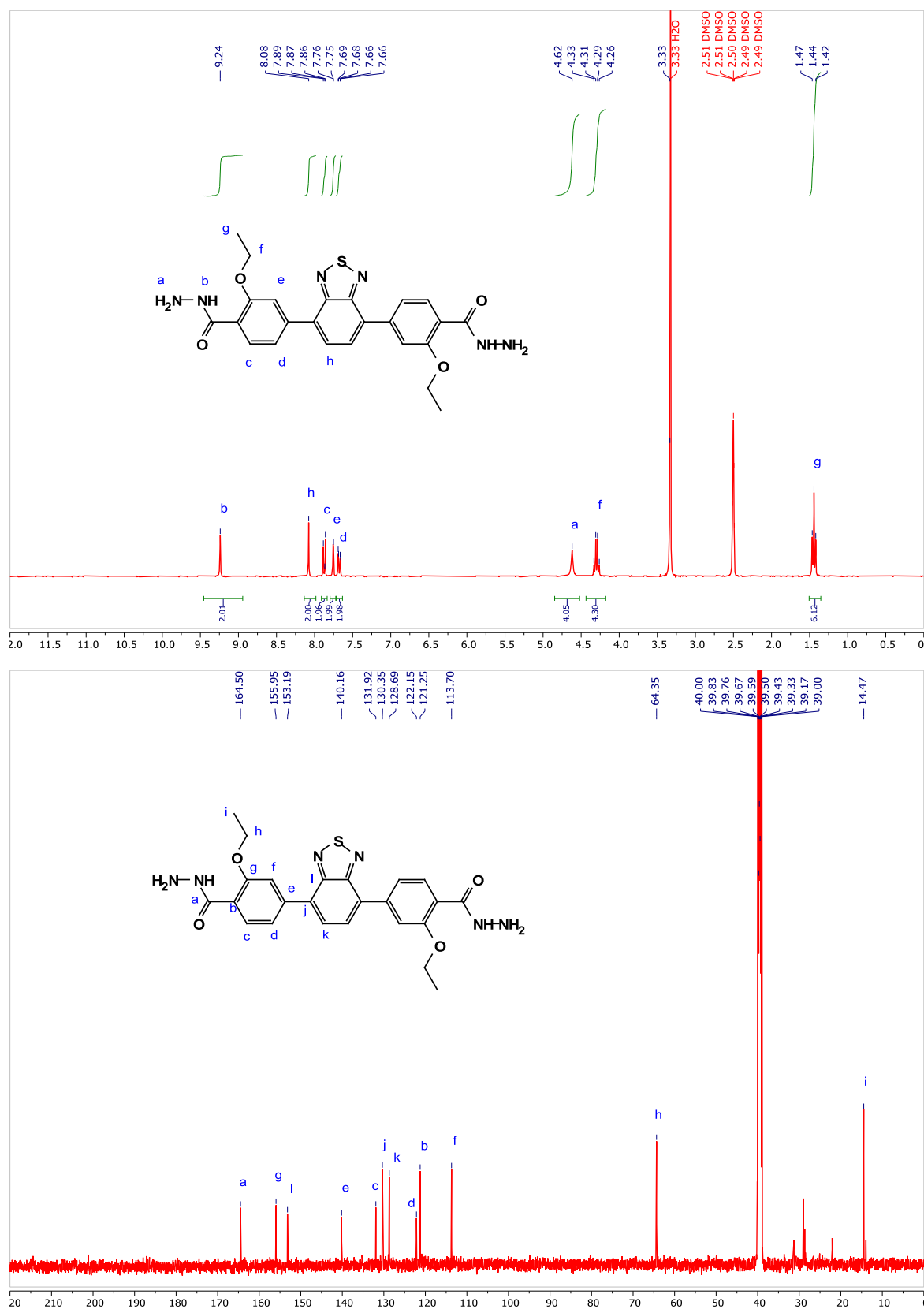
Supplementary Fig. 44 | ¹H NMR and ¹³C NMR spectra of methyl 2-ethoxy-4-iodobenzoate (C).



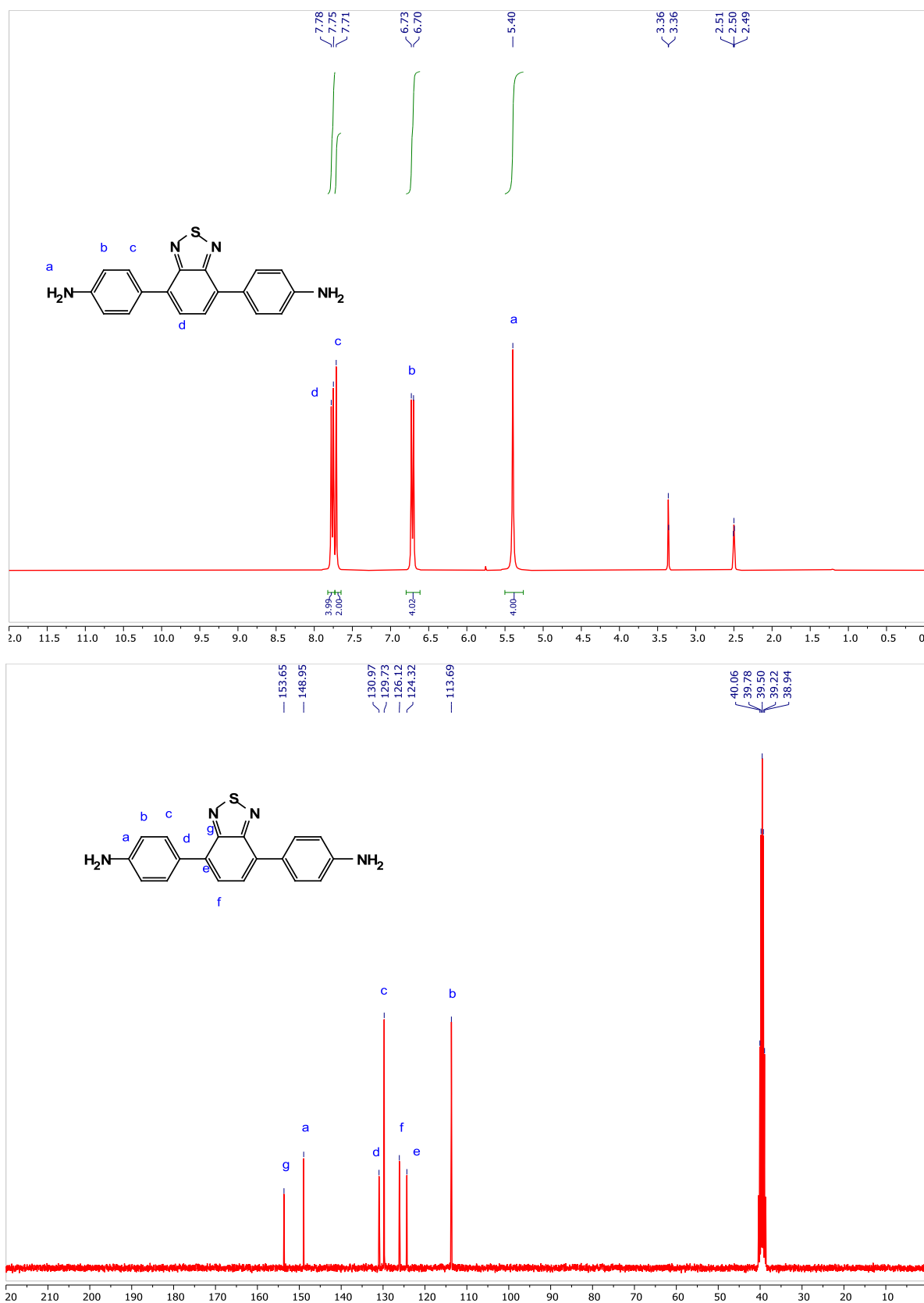
Supplementary Fig. 45 | ¹H NMR and ¹³C NMR spectra of methyl 2-ethoxy-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)benzoate (D).



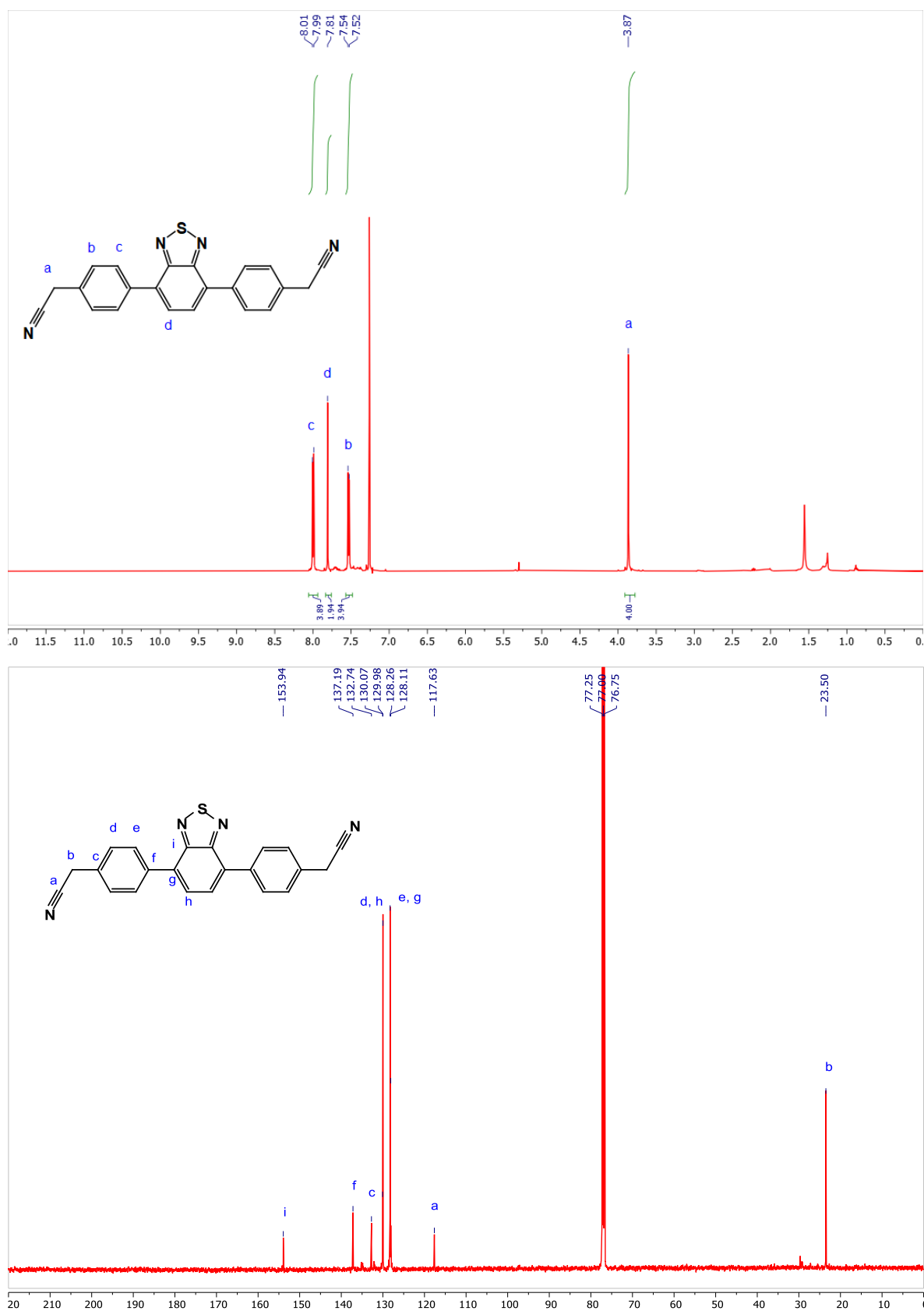
Supplementary Fig. 46 | ¹H NMR and ¹³C NMR spectra of dimethyl 4,4'-(benzo[c][1,2,5]thiadiazole-4,7-diyl)bis(2-ethoxybenzoate) (E).



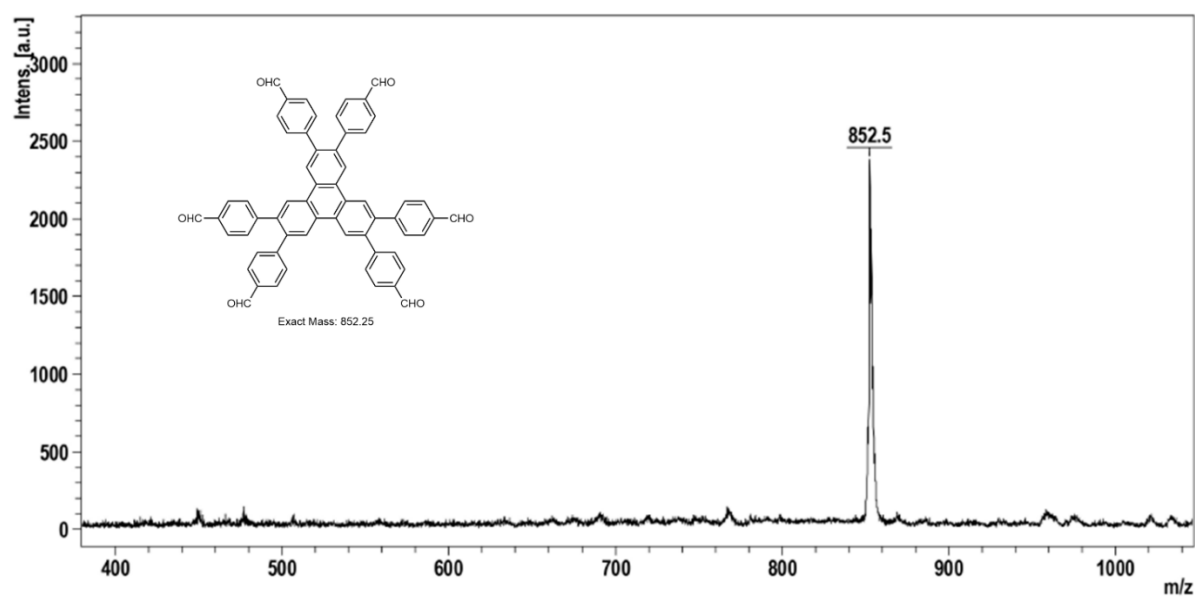
Supplementary Fig. 47 | ¹H NMR and ¹³C NMR spectra of 4,4'-(benzo[c][1,2,5]thiadiazole-4,7-diyl)bis(2-ethoxybenzohydrazide) (**F**).



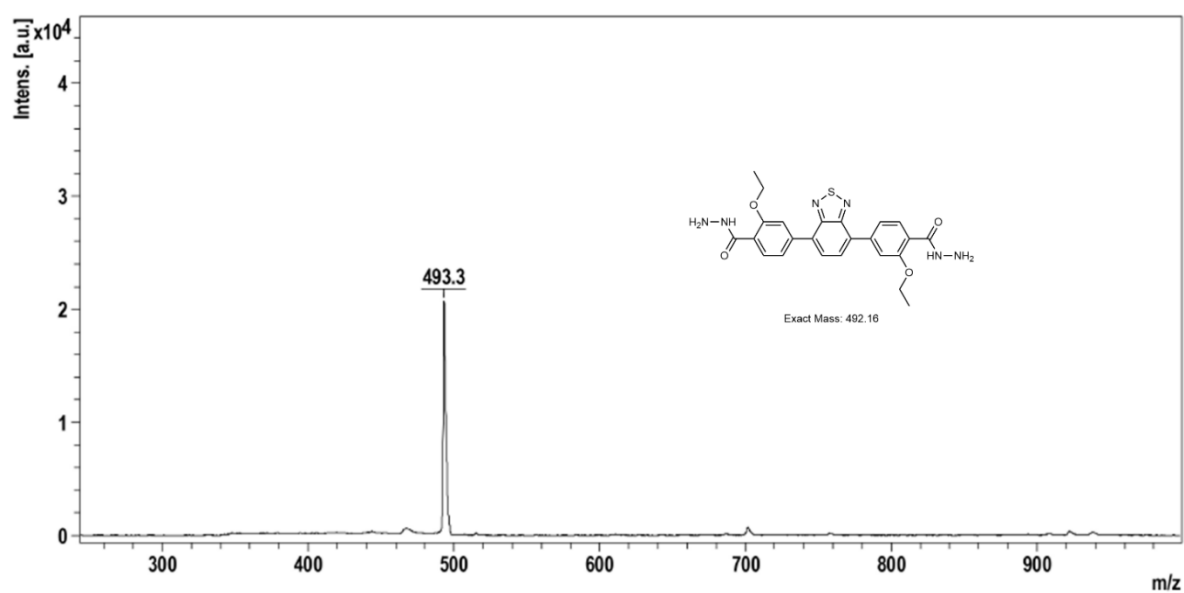
Supplementary Fig. 48 | ¹H NMR and ¹³C NMR spectra of 4,4'-(benzo[c][1,2,5]thiadiazole-4,7-diyl)dianiline (G).



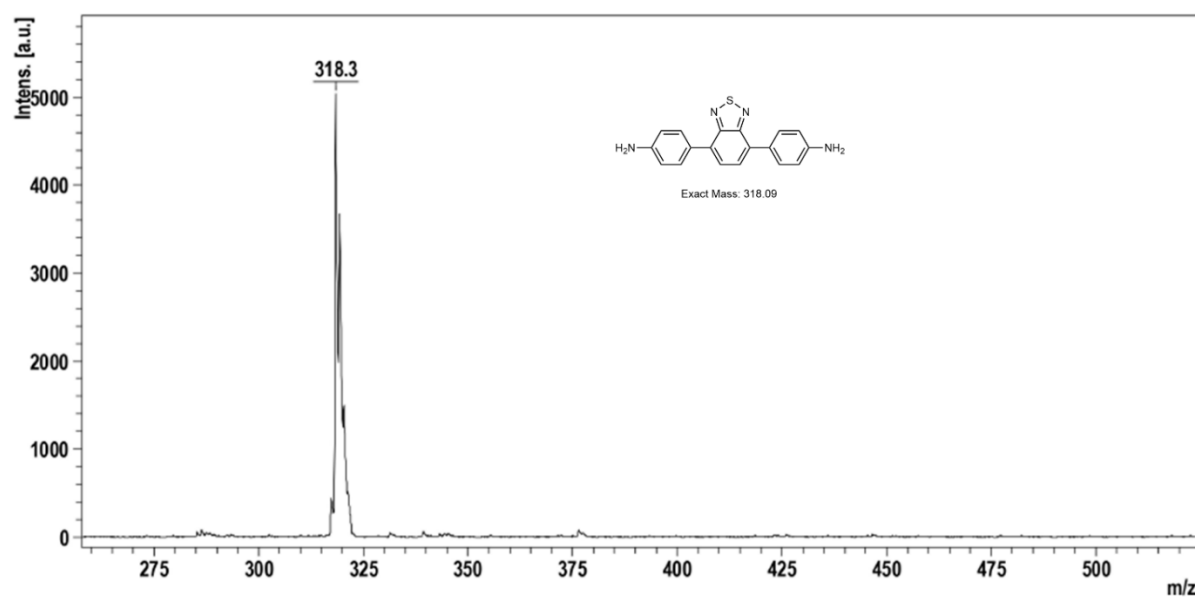
Supplementary Fig. 49 | ^1H NMR and ^{13}C NMR spectra of 2,2'-(benzo[c][1,2,5]thiadiazole-4,7-diylbis(4,1-phenylene))diacetonitrile (H).



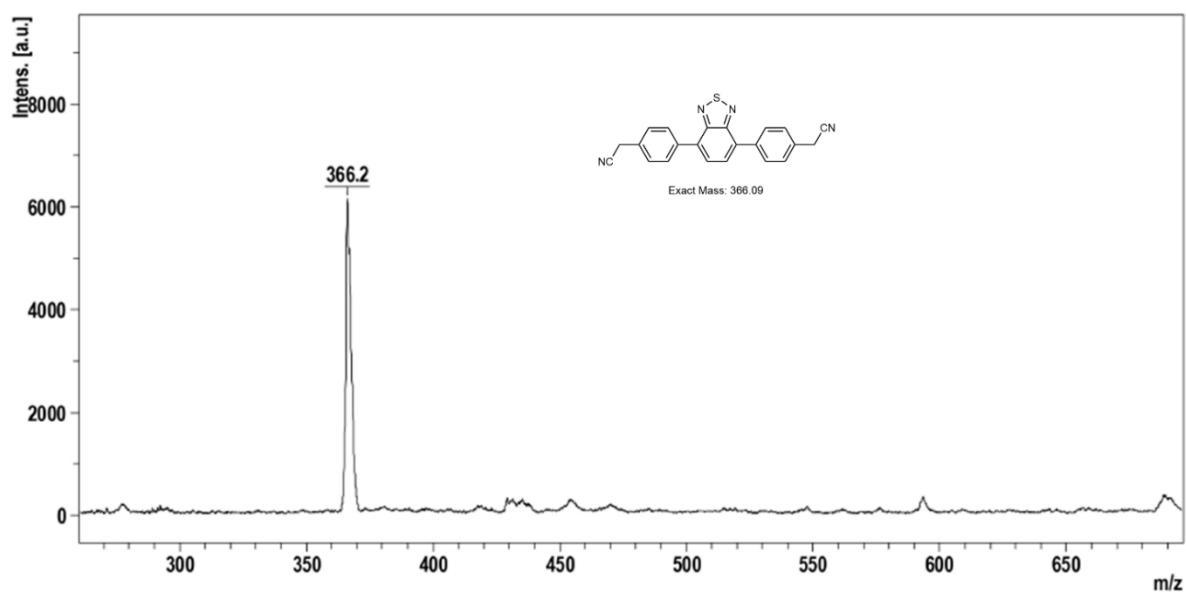
Supplementary Fig. 50 | MALDI-TOF spectra of 2,3,6,7,10,11-hexakis(4-formylphenyl)triphenylene (B).



Supplementary Fig. 51 | MALDI-TOF spectra of 4,4'-(benzo[c][1,2,5]thiadiazole-4,7-diyl)bis(2-ethoxybenzohydrazide) (F).



Supplementary Fig. 52 | MALDI-TOF spectra of 4,4'-(benzo[c][1,2,5]thiadiazole-4,7-diyl)dianiline (G).



Supplementary Fig. 53 | MALDI-TOF spectra of 2,2'-(benzo[c][1,2,5]thiadiazole-4,7-diylbis(4,1-phenylene))diacetonitrile (H).

Supplementary Discussion

Stacking modes

In addition to sync AA stacking mode, we constructed and optimised (using DFTB+) the reversed AA, sync AB and reversed AB stacking modes (Supplementary Fig. 2) and simulated their PXRD patterns (Supplementary Fig. 3).

Comparison of the simulated PXRD patterns reveal that for all three COFs, the sync AA and reversed AA stacking modes could match with the experimentally obtained PXRD patterns, in terms of peak position and relative peak intensities. However, the reversed AA stacking mode produces COFs with a singular pore size, due to its alternately stacked “small pore-large pore” layer. On the other hand, the sync AA stacking mode produces COFs with two distinct and unique pore sizes, consistent with experimentally obtained pore size measurements (Fig. 2e, Supplementary Fig. 9). In addition, we have also estimated the average pore dimensions (average distance from one benzothiadiazole to the adjacent) of the sync AA and reversed AA stacking modes. As can be seen, the difference in pore dimensions for sync AA is about 0.2 nm, while negligible differences were observed for reversed AA. Hence, the sync AA stacking mode is proposed to be the most probable.

Experimental and theoretical optoelectronic properties

As shown in Supplementary Fig. 12 and 15, from DFT calculations, Hz-TP-BT-COF (Supplementary Fig. 15a, green line) has a direct electronic band gap of 1.55 eV as calculated using the Heyd-Scuseria-Ernzerhof (HSE06) hybrid functional⁷, while Im-TP-BT-COF (orange line) and sp²c-TP-BT-COF (blue line) show indirect band gaps of 2.17 and 2.11 eV, respectively. In addition, both Im-TP-BT-COF and sp²c-TP-BT-COF exhibit substantial intralayer dispersion due to the conjugated linkages of imine and cyano-vinylene, while Hz-TP-BT shows minimal dispersion along the in-plane directions as a result of the non-conjugated hydrazine linkage, implying localised electronic structures. The photo-induced charge carriers, however, can move along out-of-plane direction, where bandwidths of 0.09 eV and 0.11 eV are observed for the highest valence band and the lowest conduction band, respectively. This is because the donor (D) knot and acceptor (A) edge linker are stacked into separate one-dimensional channels for hole and electron transport^{8,9}. Owing to the DA design principle, photoexcited electrons and holes in Hz-TP-BT-COF exhibit charge-transfer characteristics, with the valence band maximum (VBM) and conduction band minimum (CBM) located on the electron-donating node and electron-withdrawing BT edge units, as manifested by the partial charge density distribution in Fig. 3. This spatial separation of holes and electrons at the VBM and CBM promotes electron-hole pair dissociation and facilitates exciton utilisation. Hz-TP-BT-COF also has the lowest LUMO level, while both Hz-TP-BT-COF and sp²c-TP-BT-COF have similar HOMO levels. Nevertheless, all three COFs possess sufficient overpotential for one electron ORR (-4.49 eV) but only Hz-TP-BT-COF and sp²c-TP-BT-COF can promote concurrent OER (-5.67 eV). Additionally, band structure analysis (Supplementary Fig. 12) also suggested that Hz-TP-BT-COF has the lowest CBM, as well as the smallest band gap, a result approximately consistent with DFT calculations. In lieu of the large discrepancy between DFT and experimental results for Hz-TP-BT-COF (Supplementary Fig. 15b), we further performed TD-DFT calculations to obtain their absorption spectra (Supplementary Fig. 10). As shown in Supplementary Fig. 10, the calculated absorption spectra of these systems reveal that all of these COFs exhibit optical absorption in the visible and ultraviolet regions, with absorption

edges at approximately 500, 550 and 600 nm for Hz-TP-BT-COF, Im-TP-BT-COF and sp²c-TP-BT-COF, respectively. Among them, sp²c-TP-BT-COF has the smallest optical band gap of 2.03 eV, resulting from its fully conjugated linkage and strong electron-withdrawing cyano group, which lowers the LUMO energy of the linker unit. Conversely, Hz-TP-BT-COF displays the largest optical band gap of 2.48 eV, closely aligning with the experimental value of 2.59 eV. This optical band gap is notably larger than the electronic band gap of 1.55 eV, implying that the lowest-energy transition from VBM to CBM is forbidden due to the non-conjugated linkage. In comparison, Im-TP-BT-COF and sp²c-TP-BT-COF do not exhibit this restriction. Hence from these observations, it can be noted that DFT calculations do not take into account forbidden transitions in non-conjugated COFs, thus not allowing an accurate description of these COFs, eventually resulting in inconsistent HOMO and LUMO levels with experimentally determined data.

To better represent the electronic levels of Hz-TP-BT-COF, we combined results from TD-DFT and re-examined its band structure (Supplementary Fig. 15a). Noticeably, there exists an isolated band at around $E-E_f$ of 1.1 eV, where we consider to be a forbidden band for VBM photoexcitation. Hence, the more ideal CBM should be the band close to 2 eV, which yields a band gap that is more consistent with TD-DFT results. On the other hand, Im-TP-BT-COF and sp²c-TP-BT-COF exhibit a continuous conduction band. Hence, we consolidated the results from these theoretical methods and re-illustrated the HOMO, LUMO and energy gaps in Supplementary Fig. 15c. Using this new approach, Hz-TP-BT-COF has the largest energy gap (2.48 eV), as well as the highest LUMO level to drive one electron ORR, consistent with experimental results (Supplementary Fig. 15b, green line).

Interestingly, for Im-TP-BT-COF and sp²c-TP-BT-COF, although both DFT and TD-DFT suggested a smaller energy gap for sp²c-TP-BT-COF, and a higher LUMO level for Im-TP-BT-COF, the experimental results prove otherwise. We performed a literature search and also found other COF photocatalytic systems (imine linkages vs analogous vinylene linkages) that exhibit identical trend as our experimental results^{10–12}. The calculated bandgap structures are based on ideal crystal lattices with boundary conditions set in the DFTB+, which can treat large numbers of atoms but cannot give precise estimation of HOMO and LUMO levels. On the other hand, the experimental results are based on electrochemical and optical measurements, which include both defects of lattices and boundaries of COFs particles, leading to bigger bandgaps and different energy levels as compared to theoretical ones. This effect is not limited to our present COFs and similar deviation phenomena are observed for many other COF materials.

pH changes during photocatalytic H₂O₂ production

We monitored the pH change throughout the 60-min reaction using MQuant® pH-indicator strips (Photocatalytic H₂O₂ generation from Hz-TP-BT-COF). Starting at about pH 5 in air-saturated tap water (due to dissolved CO₂), pH of the solution decreases slowly towards pH 4 due to the partial dissociation of as-synthesized H₂O₂ (Supplementary Fig. 18).

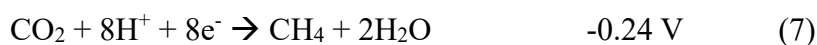
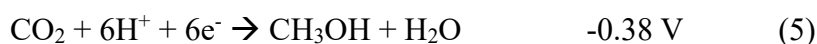
Selectivity and byproduct analysis

Taking into account that the photocatalytic system only requires air to produce H₂O₂ efficiently, it is important to rule out the possibilities of other byproducts. Considering that air is made of 78% N₂, 21% O₂, 0.9% Ar and 0.04% CO₂ (in our case, dissolved in H₂O), we went on to verify the possibility of other common photocatalytic reactions.

N₂: Although N₂ is present in the greatest abundance, photocatalytic N₂ activation is of immense challenge due to the extremely strong N₂ triple bond. Furthermore, 6-electron reduction to NH₃ is often kinetically sluggish and is often outcompeted by more favourable hydrogen evolution^{13, 14}. Besides, N₂ also has poor solubility in water. In fact, it is even more challenging to achieve this in the absence of any metal co-catalysts, without sacrificial donors.

CO₂: CO₂ is present in a minute percentage in air. To detect possible byproducts, we performed direct headspace sampling of a photoirradiated sample of Hz-TP-BT-COF, dispersed in water, under closed air environment. However, no gaseous reduction products of CO₂ (CH₄ and CO) were detected by GC (with TCD detector) (Supplementary Fig. 20a) after 6 hours of continuous irradiation. To increase the detection limit, we also photoirradiated a sample of Hz-TP-BT-COF dispersed in CO₂-saturated aqueous solution. Yet, no gaseous reduction products were detected after 6 hours (Supplementary Fig. 20b). To rule out any reduction products in the aqueous phase (e.g. CH₃OH, HCOOH), we also obtained the ¹H NMR (with water suppression) of the aqueous phase (800 μL aqueous phase + 100 μL D₂O + 50 μL *d*₆-DMSO standard) but no other carbon-containing product peaks were observed (Supplementary Fig. 21). These results suggest that photoreduction of CO₂ is not likely to proceed under the above conditions. Generally speaking, carbon dioxide photoreduction is favoured in the presence of chelated transition metal species such as Re, Ni, Zn and Co to accumulated electrons, bind and activate CO₂; organic solvents such as acetonitrile to enhance COF dispersion and carbon dioxide solubility; sacrificial donors such as triethanolamine (TEOA), 1-benzyl-1,4-dihydronicotineamide (BNAH), 1,3-dimethyl-2-phenyl-2,3-dihydro-1H-benzo[d]imidazole (BIH) to drive the oxidation half-reaction; and additional photosensitizers such as [Ru(bpy)₃]Cl₂ and (Ir[dF(CF₃)ppy]₂(dtbpy))PF₆¹⁵⁻²¹. Based on the above evidence, photoreduction of CO₂ is not likely to occur under our reaction conditions.

H₂: Hydrogen evolution is often considered as the most competitive side reaction for photocatalytic systems in aqueous solutions. To determine if any H₂ is produced, we performed direct headspace sampling of a photoirradiated sample of Hz-TP-BT-COF, dispersed in water, under closed argon environment. However, no H₂ was detected even after 6 hours of continuous irradiation (Supplementary Fig. 20c). These results suggest that hydrogen evolution is not likely to proceed under the above conditions. Generally speaking, hydrogen evolution reaction is favoured in the presence of Pt co-catalyst to accumulate electrons and serve as active site for the reaction and sacrificial donors such as ascorbic acid, triethanolamine (TEOA) and lactic acid to complete the oxidation half-reaction²²⁻²⁸. Based on the above evidence, hydrogen evolution is not likely to occur under our reaction conditions.



Besides the above experiments, kinetics and thermodynamics (Supplementary Fig. 19) also play essential roles in controlling the selectivity of reaction. As can be seen, superoxide radical formation (Supplementary Equation 6) is a one-electron process, which occurs at a faster rate

than the other reactions involving two to eight electrons. Compared to other reduction reactions which occurs at lower potential (Supplementary Equations 2 to 5), there is a greater overpotential for superoxide radical formation. On the other hand, compared to the eight-electron CO_2/CH_4 reaction (Supplementary Equation 7), the one-electron superoxide radical formation is inevitably easier to achieve. Therefore, superoxide radical formation is deemed the most favourable under our conditions.

Finally, as mentioned in the main text, Hz-TP-BT-COF is stable under various conditions, even tolerating extended usage under flow reactor for at least 144 hours, without structural degradations. This also indicates that the radical intermediates produced is fully utilised in H_2O_2 production and not consumed in side-reactions with the photocatalyst. These experiments further highlight the efficiency of our system, where a superior turnover frequency is achieved even under lower oxygen levels, in the presence of other competing species.

ICP-MS results to measure Pd content

Palladium could be the most likely source of contamination due to its use as a catalyst in cross-coupling reactions in monomer synthesis. To eliminate the possibility of any metal contamination, we performed inductively coupled plasma mass spectroscopy on Hz-TP-BT-COF before and after the photocatalytic reaction. Nonetheless, Pd was not detected in both samples. This suggests that the high efficiency of Hz-TP-BT-COF in photocatalytic H_2O_2 production is not due to metal contamination. Besides, metal-free photocatalytic H_2O_2 production has been reported across covalent organic frameworks, conjugated polymers, covalent triazine frameworks, resorcinol formaldehyde resins etc (Supplementary Table 7). Hence, the likelihood of such reactions being manipulated by metal contaminants is low.

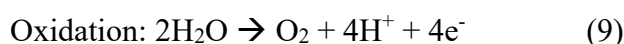
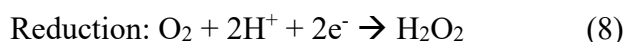
Gibbs free energy

H_2O_2 generation involves a two-electron reduction of O_2 ^{29, 30}. For each COF, both BT and Ph on the linker units are considered as reaction sites (Fig. 6b and 6c, Extended Data Fig. 4a) due to the predominant localisation of the photogenerated electron on the edge (Fig. 3). The reaction mechanism is as follows: initially, the COF photocatalysts transfer the photogenerated electron to the molecular oxygen, producing a superoxide radical anion that combines with H^+ in the solution to form the HOO^* intermediate. Subsequently, a second proton-coupled-electron-transfer reaction occurs to generate H_2O_2 (Extended Data Fig. 4b). As shown in Fig. 6a, $\Delta G_1(\text{BT})$ is generally smaller than $\Delta G_1(\text{Ph})$ on all the COFs, suggesting that BT is more active. Among them, the BT units of Hz-TP-BT-COF has the smallest ΔG_1 of 0.50 eV, while Ph also shows a comparable ΔG_1 of 0.56 eV, indicating both are active for H_2O_2 production. This finding explains the superior photocatalytic H_2O_2 generation performance of Hz-TP-BT-COF compared to Im-TP-BT-COF and sp^2c -TP-BT-COF in the experiment. Regarding the four-electron OER, both single-site and dual-site pathways for the Hz-TP-BT-COF on the neighbouring Ph unit of the triphenylene knot are calculated (Extended Data Fig. 4c). As depicted in the energy profile in Fig. 6e, the first step of forming $^*\text{OH}$ has the largest ΔG of 1.76 eV for the dual-site pathway, while the third step of forming $^*\text{OOH}$ is the rate-determining step for the single-site pathway. This finding suggests that the dual-site mechanism may be more energetically favourable than the single-site one^{31, 32} (Extended Data Fig. 4d).

Control experiments to quantify O₂ production during H₂O₂ generation

Typically, previous reports testify and quantify O₂ production (alongside H₂O₂ production) by the addition of sacrificial acceptors such as AgNO₃ or NaIO₃; co-catalysts such as Co²⁺ and La₂O₃ as pH buffer^{33–35}. However, the addition of these substances inevitably alters the reaction mechanism in favour of oxygen evolution reaction, thus not reflecting the true quantity of oxygen produced.

The photocatalytic H₂O₂ production in our system has been proven to involve two half-reactions (Supplementary Equations 8 and 9):



Although the oxidation half-reaction (Supplementary Equation 9) produces oxygen, the same oxygen is consumed (together with oxygen from input) via reduction to form H₂O₂ (Supplementary Equation 8). To prove this point, we performed photocatalytic H₂O₂ production with Hz-TP-BT-COF, using H₂¹⁸O as an isotope and performed GC-MS analysis of the headspace after photoirradiation (Supplementary Fig. 40). As can be seen, the peak belonging to ¹⁸O₂ was detected, suggesting that oxygen evolution (Supplementary Equation 9) indeed occurred. Thereafter, we purged the reaction with argon, extracted the aqueous phase, removed the photocatalyst, before injecting it into a vial containing MnO₂ (oxidises H₂O₂ to O₂). We then performed headspace sampling and GC-MS analysis, which revealed the formation of ¹⁸O₂ (Supplementary Fig. 41). This provides evidence that the oxygen produced in (9) is used for the formation of H₂O₂. Similar methods were also adopted in other papers^{33–37}.

After confirming that oxygen is produced, and simultaneously consumed, we can start to examine Supplementary Equation (10). As can be seen, due to the great abundance of H₂O, O₂ is the limiting reagent. Hence, if the reaction is conducted in an air-tight environment, we would expect O₂ to be consumed gradually, while H₂O₂ concentration rises at a decreasing rate. To verify the above hypothesis, we dispersed Hz-TP-BT-COF in pre-oxygen saturated water, in an air-tight vessel. After continuous photoirradiation for 3 hours, we found that the H₂O₂ concentration rose to a plateau at about 0.48 mM and no additional production was observed for the next hour even with photoirradiation. This value corresponds to approximately half of the solubility of O₂ in water (0.25 mM)³⁸, which is close to the stoichiometric amount (1:2) from Supplementary Equation (10). Subsequently, we re-introduced oxygen into the reaction vessel and continued photoirradiation, and the concentration of H₂O₂ was observed to increase again (Supplementary Fig. 42). This suggests that the stop in H₂O₂ production at 3 hours is due to depleted O₂ in the closed reaction environment. From the above experiment, we are able to simultaneously correlate H₂O₂ production with O₂ depletion, and further verify the mechanism for our system.

Supplementary Tables

Supplementary Table 1 | Unit cell parameters and Pawley refinement *R*-factors of COFs.

	Hz-TP-BT-COF	Im-TP-BT-COF	sp ² c-TP-BT-COF
<i>a</i> / Å	38.67086	34.23001	34.38256
<i>b</i> / Å	38.85580	34.53142	34.47077
<i>c</i> / Å	3.67443	3.48600	3.68680
α / °	90	90	90
β / °	90	90	90
γ / °	60	60	60
Space group	<i>P</i> 1	<i>P</i> 1	<i>P</i> 1
<i>R</i> _{wp} / %	3.80	3.65	2.94
<i>R</i> _p / %	2.87	2.50	2.31

Supplementary Table 2 | Fitting parameters and lifetimes for the three COFs.

Sample	<i>A</i> ₁	τ_1 (ps)	<i>A</i> ₂	τ_2 (ps)	τ_{average}^* (ps)
Hz-TP-BT-COF	0.24	8.6 ± 0.2	0.22	147 ± 3	75 ± 2
Im-TP-BT-COF	0.37	1.0 ± 0.4	0.25	16.9 ± 0.5	7.4 ± 0.4
sp ² c-TP-BT-COF	0.29	1.8 ± 0.4	0.25	81 ± 3	38 ± 2

* The τ_{average} was calculated as weighted average.

Supplementary Table 3 | ICP-MS measurement to determine Pd content before and after the photocatalytic reaction.

Sample	Pd content (%)
Pristine Hz-TP-BT-COF	Not detected
Hz-TP-BT-COF after photocatalysis	Not detected

Supplementary Table 4 | Comparison of performance with reported photocatalysts.

Type	Photocatalyst	Solvent	Gas	λ (nm)	Light Source (W)	Reduc- tion	Oxida- tion	H ₂ O ₂ rate (mmol g ⁻¹ h ⁻¹)	H ₂ O ₂ conc. (mM g ⁻¹ h ⁻¹)	AQE (%)	TOF [^] (h ⁻¹)	Ref
COFs	TAPD-(Me) ₂	H ₂ O / EtOH (9:1)	O ₂	420 - 700	250	2-step ORR	Alcohol oxidation	0.1	19	N.R.	-	39
	TfpBpy	H ₂ O	air	420 - 700	300	1-step ORR	1-step WOR	0.7	69	8.1 (420 nm)	0.20	36
	TF ₅₀ -COF	H ₂ O / EtOH (9:1)	O ₂	> 400	300	2-step ORR	N.R.	1.7	35	5.1 (400 nm)	0.97	40
	TTA-TTTA	H ₂ O	O ₂	420	50	1-step ORR	OER	2.4	48	N.R.	-	41
	SonoCOF-F2	H ₂ O	air	full	1,440	2-step ORR	OER	1.2	249	4.8 (420 nm)	-	42
	N ₆ -COF	H ₂ O	O ₂	495	N.R.	2-step ORR	OER	1.6	79	N.R.	0.31	43
	EBA-COF	H ₂ O / EtOH (9:1)	O ₂	420	50	2-step ORR	Alcohol oxidation	1.8	732	4.4 (420 nm)	0.77	44
	TPB-DMTP*	H ₂ O	O ₂	> 420	300	2-step ORR	N.R.	1.6	31	18.4 (420 nm)	0.92	45
	TAPB-PDA- OH	H ₂ O / EtOH (9:1)	O ₂	> 420	10	2-step ORR	Alcohol oxidation	2.1	212	3.0 (420 nm)	-	46
	TAPT-HTA- COF	H ₂ O / IPA (9:1)	O ₂	> 420	300	2-step ORR	N.R.	1.5	49	5.4 (420 nm)	-	47
	NUST-16	H ₂ O / EtOH (9:1)	O ₂	> 420	300	2-step ORR	2-step WOR	1.1	22	N.R.	-	48
	DBTP	H ₂ O + HCl (pH 2.5)	O ₂	420 - 780	300	N.R.	OER	10.0	501	7.4 (450 nm)	-	49
	DETH-COF	H ₂ O	air	> 420	300	2-step ORR	Hydration -induced	1.0	20	0.063 (450 nm)	0.25	50
	CoPc-BTM- COF	H ₂ O / EtOH (9:1)	O ₂	> 400	300	1-step ORR	Alcohol oxidation	2.1	42	7.2 (630 nm)	2.0	51
	HEP-TAPT- COF	H ₂ O	O ₂	> 420	300	2-step ORR	OER	1.8	18	15.35 (420 nm)	0.69	33
	TTF-BT-COF	H ₂ O	O ₂	> 420	300	2-step ORR	2-step WOR	2.8	276	11.19 (420 nm)	1.09	37
	DMCR-1NH	H ₂ O / IPA (10:1)	O ₂	420 - 700	300	2-step ORR	Alcohol oxidation	2.6	129	10.2 (420 nm)	-	52
	TaptBtt	H ₂ O	Air	> 420	300	2-step ORR	1-step WOR	1.4	141	4.6 (450 nm)	0.22	53
	TpDz	H ₂ O	O ₂	> 420	300	1-step ORR	OER	7.3	407	11.9 (420 nm)	0.65	54
	TpDz	H ₂ O	Air	> 420	300	1-step ORR	OER	4.6	253	N.R.	0.40	54
	TZ-COF	H ₂ O	Air	> 420	300	2-step ORR	OER	0.3	8.9	0.6 (475 nm)	0.055	55
	FS-COFs	H ₂ O	O ₂	> 420	300 + 10	1-step ORR	OER	3.9	195	6.21 (420 nm)	1.9	56

(This table continues to the next page)

Type	Photocatalyst	Solvent	Gas	λ (nm)	Light Source (W)	Reduction	Oxidation	H ₂ O ₂ rate (mmol g ⁻¹ h ⁻¹)	H ₂ O ₂ conc. (mM g ⁻¹ h ⁻¹)	AQE (%)	TOF (h ⁻¹)	Ref
Covalent Triazine Frameworks	CTF-BDDBN	H ₂ O	O ₂	> 420	300	1-step ORR	1-step WOR	0.1	1.9	N.R.	0.025	34
	Bpt-CTF	H ₂ O	O ₂	350 - 780	300	2-step ORR	OER	3.3	65	8.6 (400 nm)	1.4	57
Covalent Heptazine Frameworks	CHF-DPDA	H ₂ O	O ₂	> 420	300	1-step ORR	1-step WOR	1.7	86	16 (420 nm)	0.32	35
Covalent Organic Polymer	AQTEE-COP	H ₂ O	O ₂	> 400	300	2-step ORR	OER	3.2	64	3.59 (400 nm)	2.4	58
Linear Polymer	DE7-M	H ₂ O	air	full	1,440	2-step ORR	OER	1.2	400	8.7 (420 nm)	-	59
Carbonaceous carbon	HTCC	H ₂ O	O ₂	> 420 nm	300	2-step ORR	N.R.	1.2	23	18.2 (420 nm)	0.077	60
g-C ₃ N ₄	DCNS	H ₂ O / EtOH (9:1)	O ₂	> 400 nm	300	N.R.	N.R.	3.1	308	6.8 (400 nm)	-	61
	ACNN	H ₂ O / IPA (9:1)	O ₂	> 420 nm	300	N.R.	N.R.	10.2	204	30.7 (429 nm)	-	62
	OCN-500	H ₂ O	O ₂	> 420 nm	300	2-step ORR	N.R.	0.1	2.0	10.2 (420 nm)	0.017	63
	Sb-SAPC15	H ₂ O	O ₂	> 420	300	2-step ORR	OER	0.6	12	17.6 (420 nm)	0.67	64
Resorcinol Formaldehyde resin	RF523	H ₂ O	O ₂	> 420 nm	2,000	N.R.	OER	0.05	1.7	~7.5 (420 nm)	-	65
	MRFS-7	H ₂ O	O ₂	> 420 nm	300	N.R.	OER	0.6	5.7	N.R.	-	66
MOFs	MIL-125-R7	H ₂ O / BnOH (2:5)	O ₂	> 420 nm	N.R.	2-step ORR	Alcohol oxidation	1.0	500	N.R.	0.21	67
	OPA/Zr _{92.5} Ti _{7.5} -MOF	H ₂ O / BnOH (2:5)	O ₂	> 420 nm	500	2-step ORR	Alcohol oxidation	3.9	1,940	N.R.	-	68
Self-Assembly	SA-TCPP	H ₂ O	O ₂	> 420	300	2-step ORR	RCOOH Oxidation	1.2	23	14.9 (420 nm)	-	69
MOC	Co ₁₄ (L-CH ₃) ₂₄	H ₂ O	O ₂	300 – 1100	300	2-step ORR	2-step WOR	0.15	15	N.R.	0.027	70
TiO ₂	Nano-BZT	H ₂ O	O ₂	> 420	300	2-step ORR	OER	1.5	29	N.R.	0.65	71
	Pd/APTMS/TiO ₂	PBS buffer	Air	AM 1.5G	N.R.	2-step ORR	OER	2.7	270	0.23 (full)	-	72
	Pt/TiO ₂	H ₂ O	-	N.R.	500	H ₂ evolution	OH ⁻ oxidation	5.1	255	N.R.	-	73
Chalcogenide	CdS NRs	H ₂ O	O ₂	Vis.	300	2-step ORR	N.R.	1.6	32	2.1 (420 nm)	-	74
Inorganic	CoO _x /Mo:BiVO ₄ /Pd	PBS buffer	O ₂	AM 1.5G	300	ORR	OER	8.6	713	5.8 (420 nm)	-	75

(This table continues to the next page)

Type	Photocatalyst	Solvent	Gas	λ (nm)	Light Source (W)	Reduction	Oxidation	H ₂ O ₂ rate (mmol g ⁻¹ h ⁻¹)	H ₂ O ₂ conc. (mM g ⁻¹ h ⁻¹)	AQE (%)	TOF (h ⁻¹)	Ref
Hybrid / Composite	SCN/T9	H ₂ O	O ₂	300 – 700	300	1-step ORR	1-step WOR	2.1	42	0.61 (365 nm)	-	76
Hybrid / Composite	PM-CDs-30	Seawater	Air	> 420	N.R.	ORR	OER	1.8	90	0.99 (420 nm)	-	77
Hybrid / Composite	SnS ₂ /In ₂ S ₃ /C Ds	H ₂ O	Air	> 420	N.R.	ORR	OER	1.1	73	3.9 (535 nm)	-	78
Hybrid / Composite	PF ₂ FBT/TiO ₂	H ₂ O	O ₂	> 420	300	ORR	OER	0.06	2.2	0.3 (460 nm)	-	79
Hybrid / Composite	Pd/A/BiVO ₄	PBS buffer	Air	> 420	300	1-step ORR	2-step WOR	0.8	8.0	6.02 (400 nm)	-	80
Hybrid / Composite	Co-CN@G	Seawater	O ₂	AM 1.5G	N.R.	2-step ORR	OER	0.4	7.1	9.1 (420 nm)	-	81
Hybrid / Composite	CN/rGO@BP QDs-0.04	H ₂ O	O ₂	420 - 780	300	N.R.	N.R.	0.06	1.2	N.R.	-	82
Hybrid / Composite	CN _{QDs} @MA-Ag (30%)	H ₂ O	O ₂	> 420	300	2-step ORR	2-step WOR	0.04	4.0	N.R.	-	83
Hybrid / Composite	MoO ₃ /SnS ₂	H ₂ O	O ₂	Solar Sim.	N.R.	2-step ORR	OER + OH ⁻ oxidation	0.8	8.3	N.R.	-	84
Hybrid / Composite	LBOB	Seawater	O ₂	427	40	2-step ORR	OH ⁻ oxidation	0.06	1.9	N.R.	-	85
Hybrid / Composite	CN _{1.8} /ICT/C Ds	H ₂ O	O ₂	> 420	N.R.	2-step ORR	2-step WOR	2.2	147	12.65 (365 nm)	-	86
COF	TPB-DMTP-COF [*]	H ₂ O	air	420 - 600	300	2-step ORR	OER	1.0	67	4.0 (420 nm)	-	This work
	Hz-TP-BT-COF	H ₂ O	air	420 - 600	300	N.R.	N.R.	5.7	378	17.5 (420 nm)	4.2	
	Hz-TP-BT-COF	H ₂ O	O ₂	420 - 600	300	N.R.	N.R.	6.5	430	-	4.8	
	Hz-TP-BT-COF	H ₂ O	air	full	300	N.R.	N.R.	12.5	830	-	9.2	
	Hz-TP-BT-COF	H ₂ O	air	sun	-	N.R.	N.R.	2.7	54	-	2.0	
Amorphous Polymer	Amorphous Hz-TP-BT polymer	H ₂ O	air	420 - 600	300	N.R.	N.R.	0.9	60	-	0.67	

*TPB-DMTP-COF was prepared and tested under identical conditions as stated.

[^]Turnover frequency is calculated based on the rate of H₂O₂ production (mmol g⁻¹ h⁻¹) divided by the amount of active site involved in H₂O₂ production per gram of photocatalyst (mmol g⁻¹). If the exact active site has not been investigated in the paper, the TOF is not calculated.

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